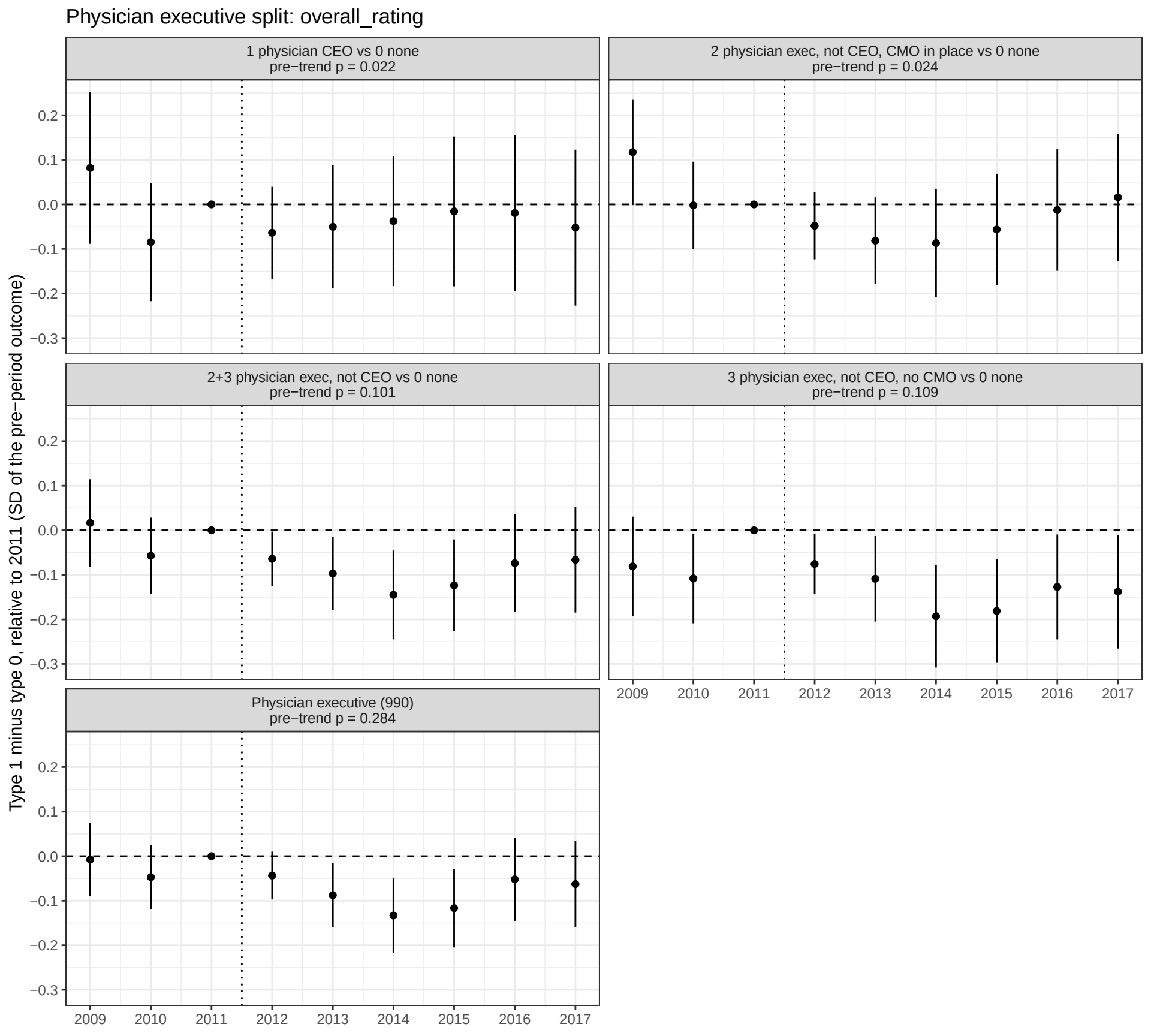
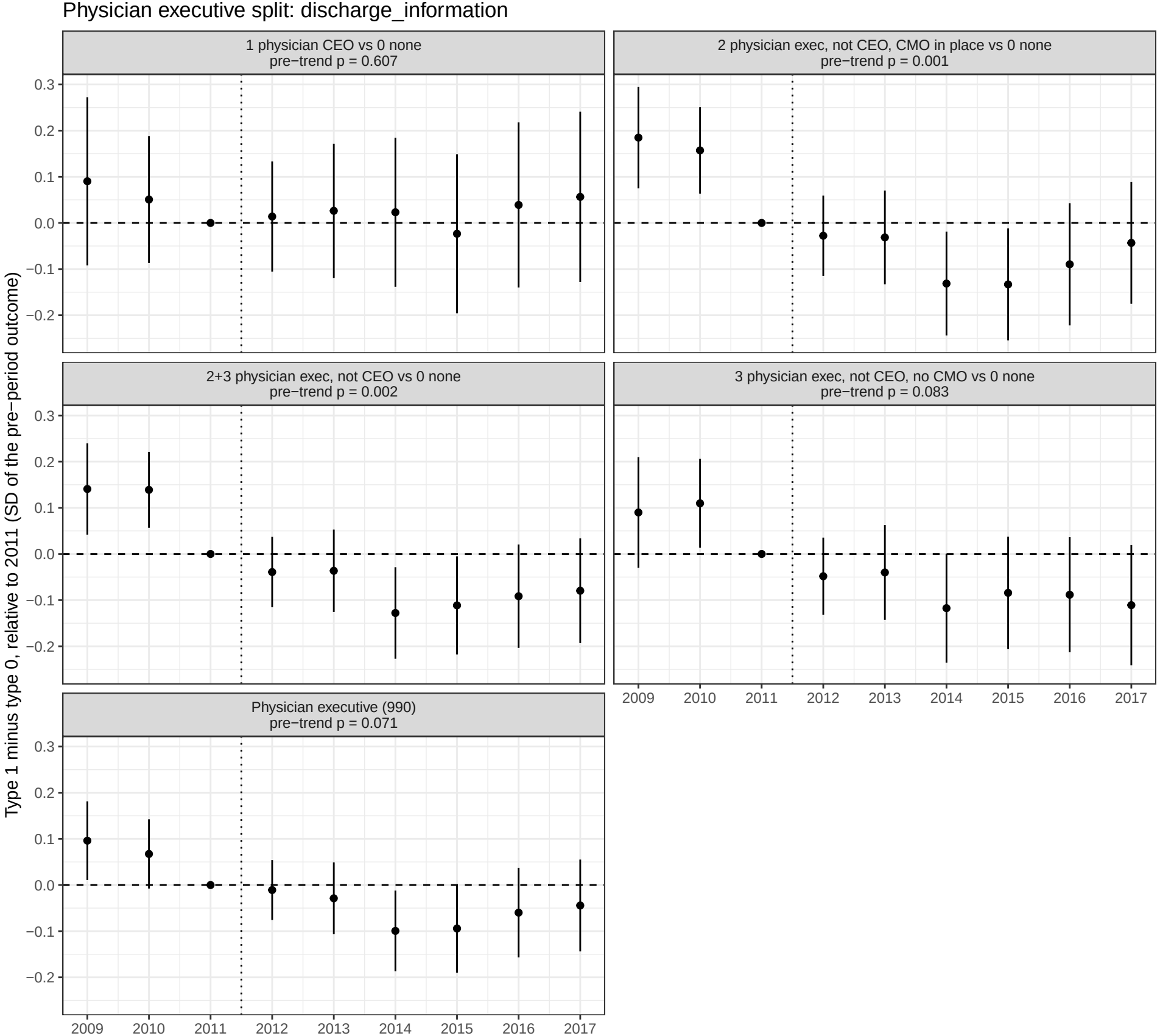
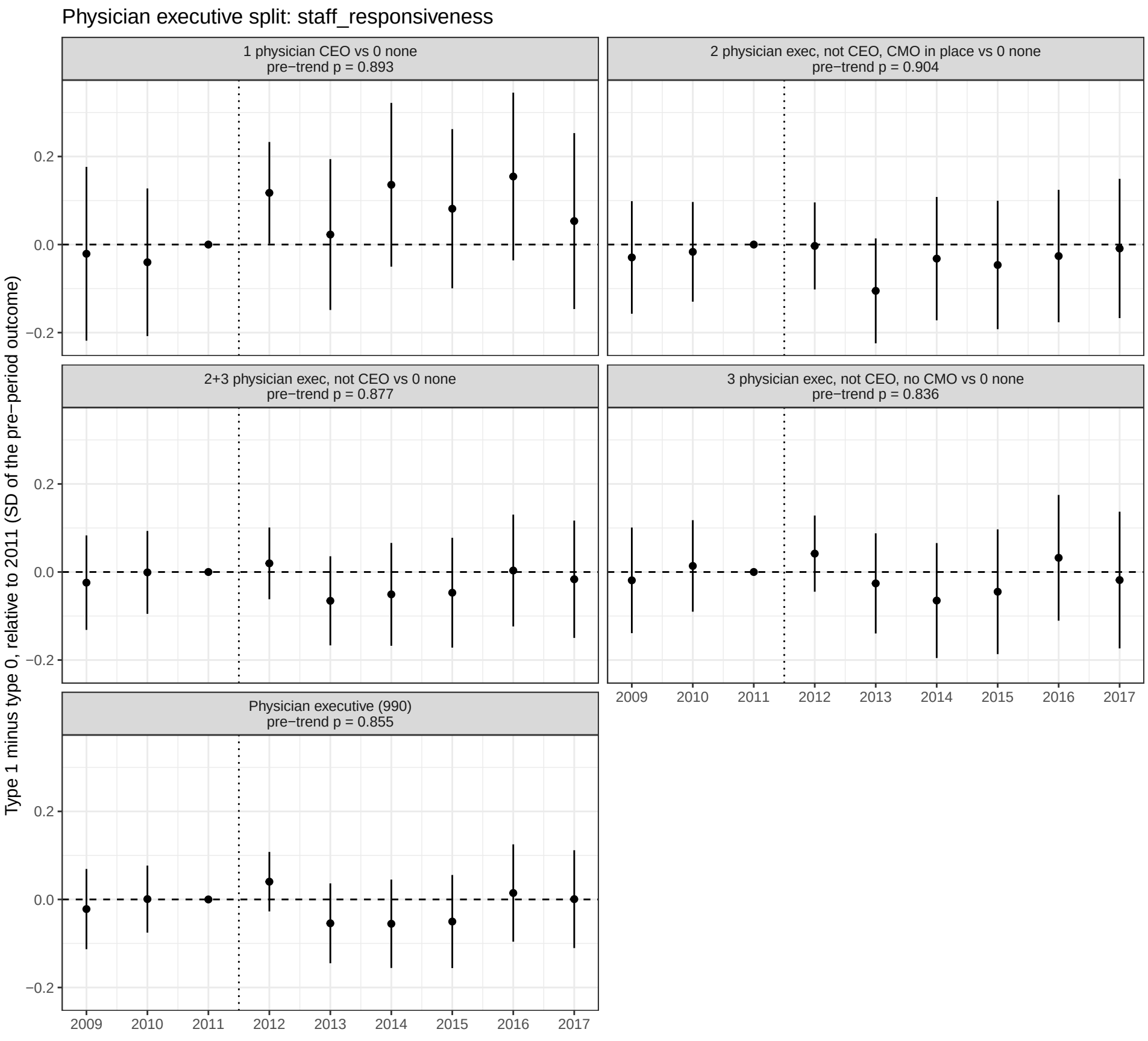
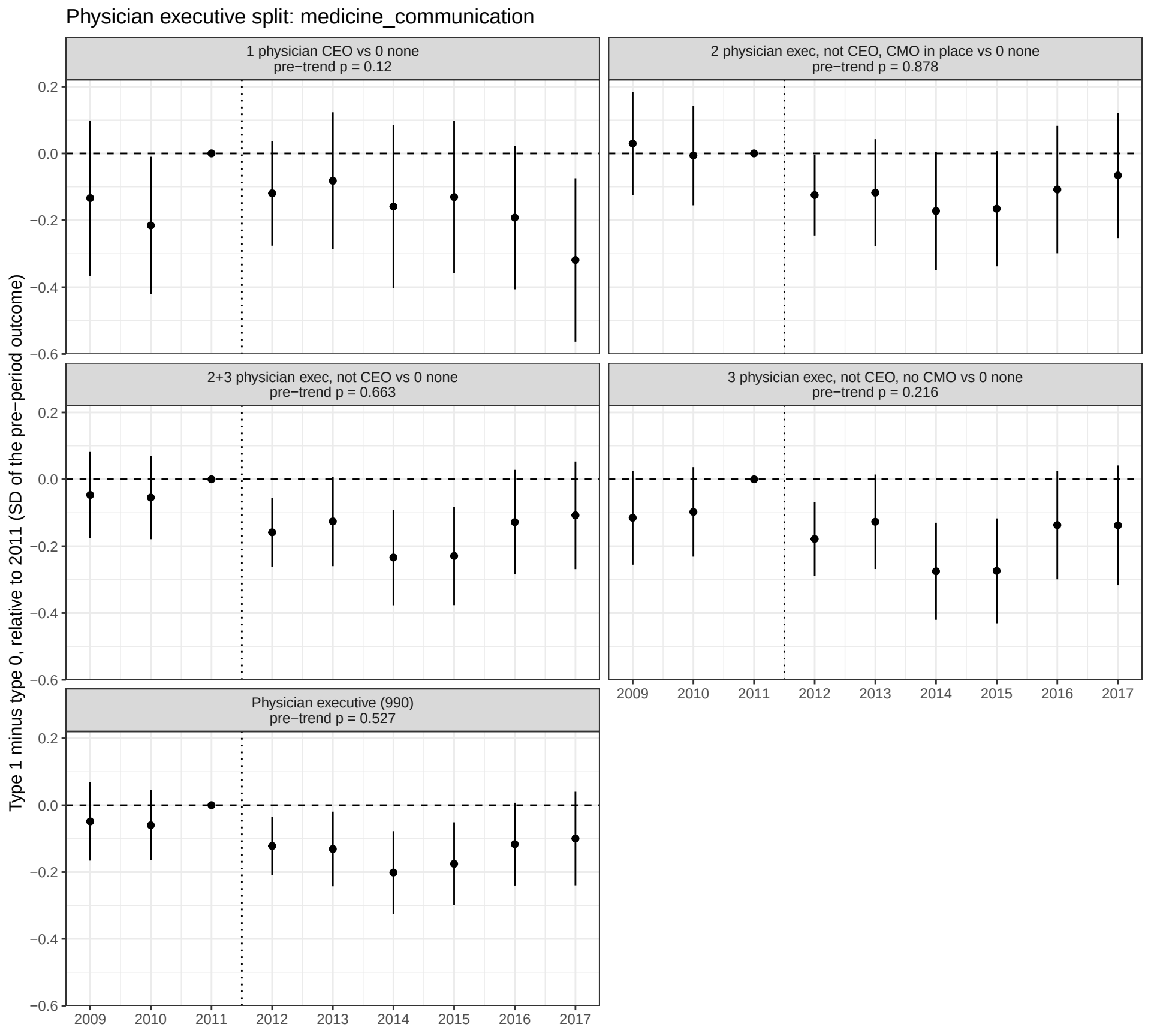
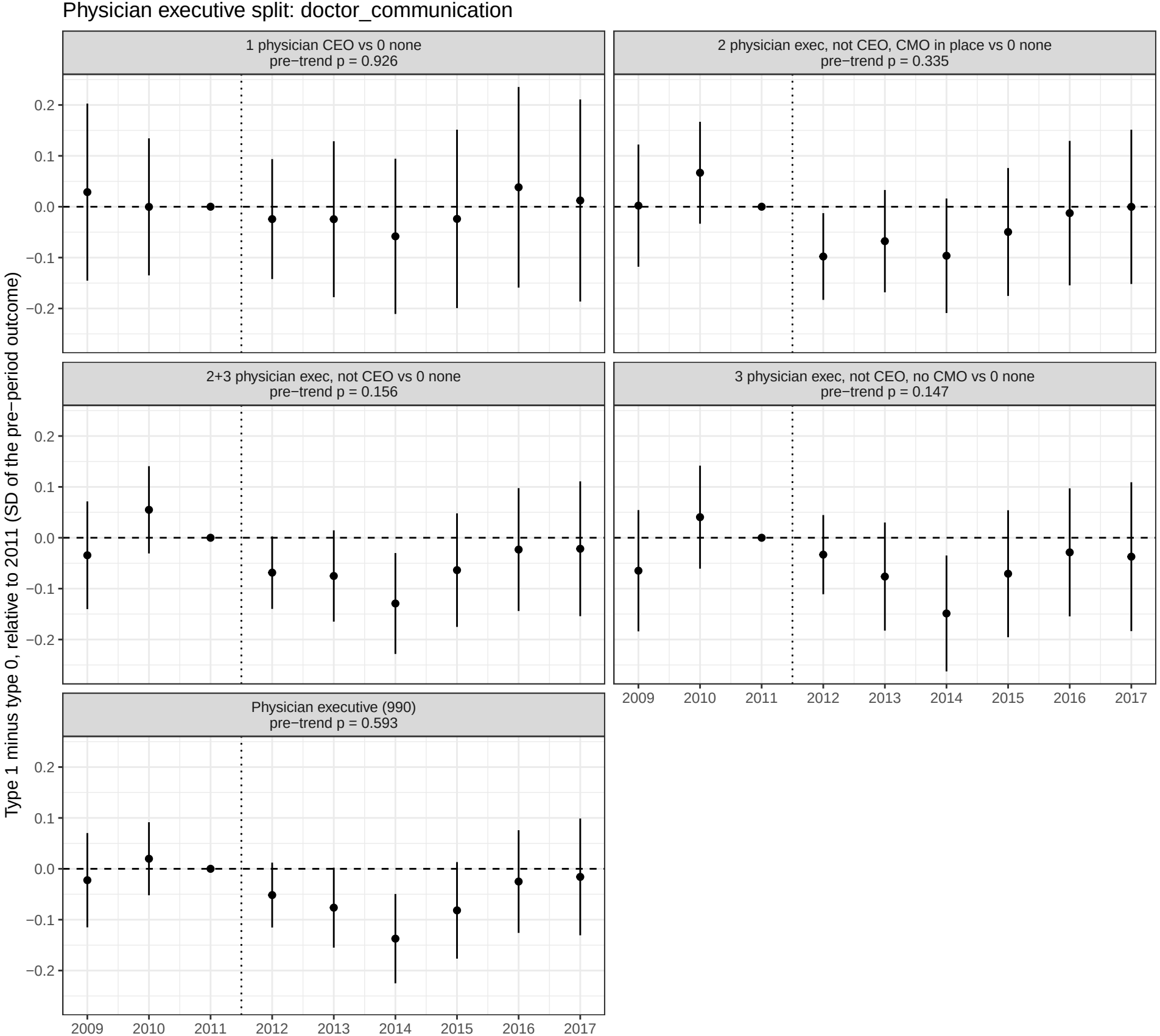
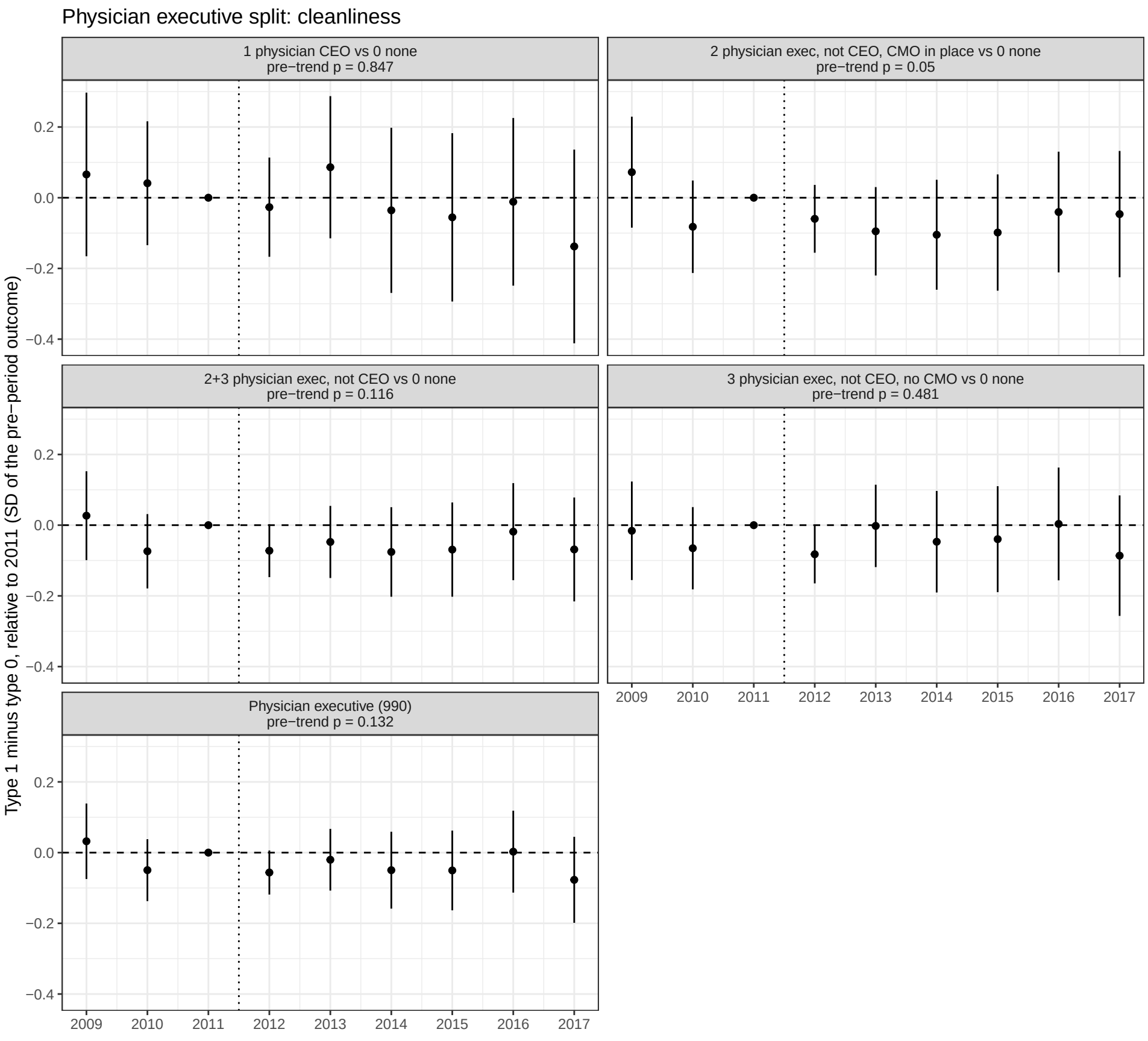

Physician executive split: overall_rating
1 physician CEO vs 0 none
pre−trend p = 0.022
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.024
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.101
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.109
Physician executive (990)
pre−trend p = 0.284
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.1
0.0
−0.1
−0.2
−0.3
2009
2010
2011
2012
2013
2014
2015
2016
2017

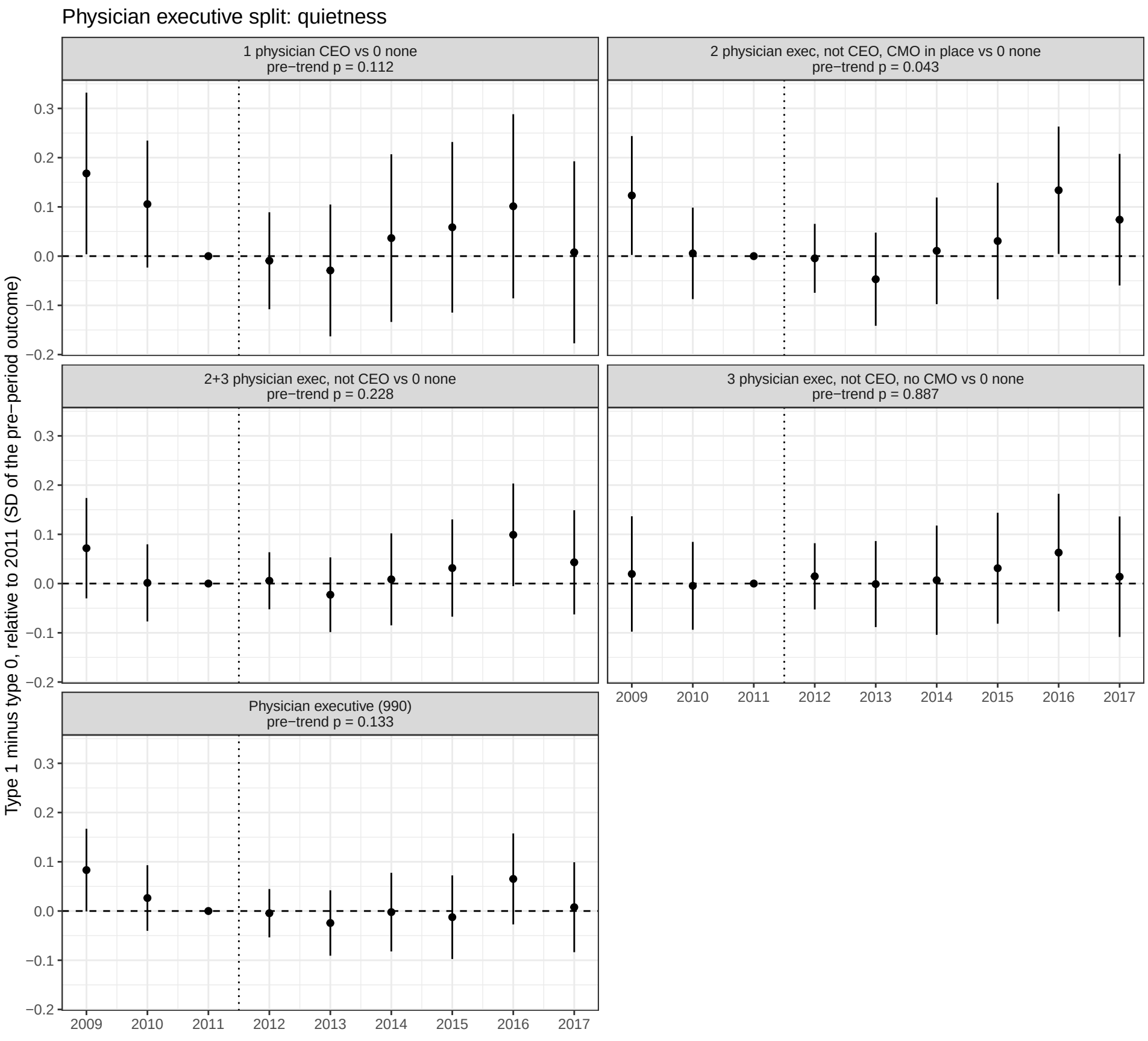

Physician executive split: discharge_information
1 physician CEO vs 0 none
pre−trend p = 0.607
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.001
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.002
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.083
Physician executive (990)
pre−trend p = 0.071
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.3
0.2
0.1
0.0
−0.1
−0.2
2009
2010
2011
2012
2013
2014
2015
2016
2017

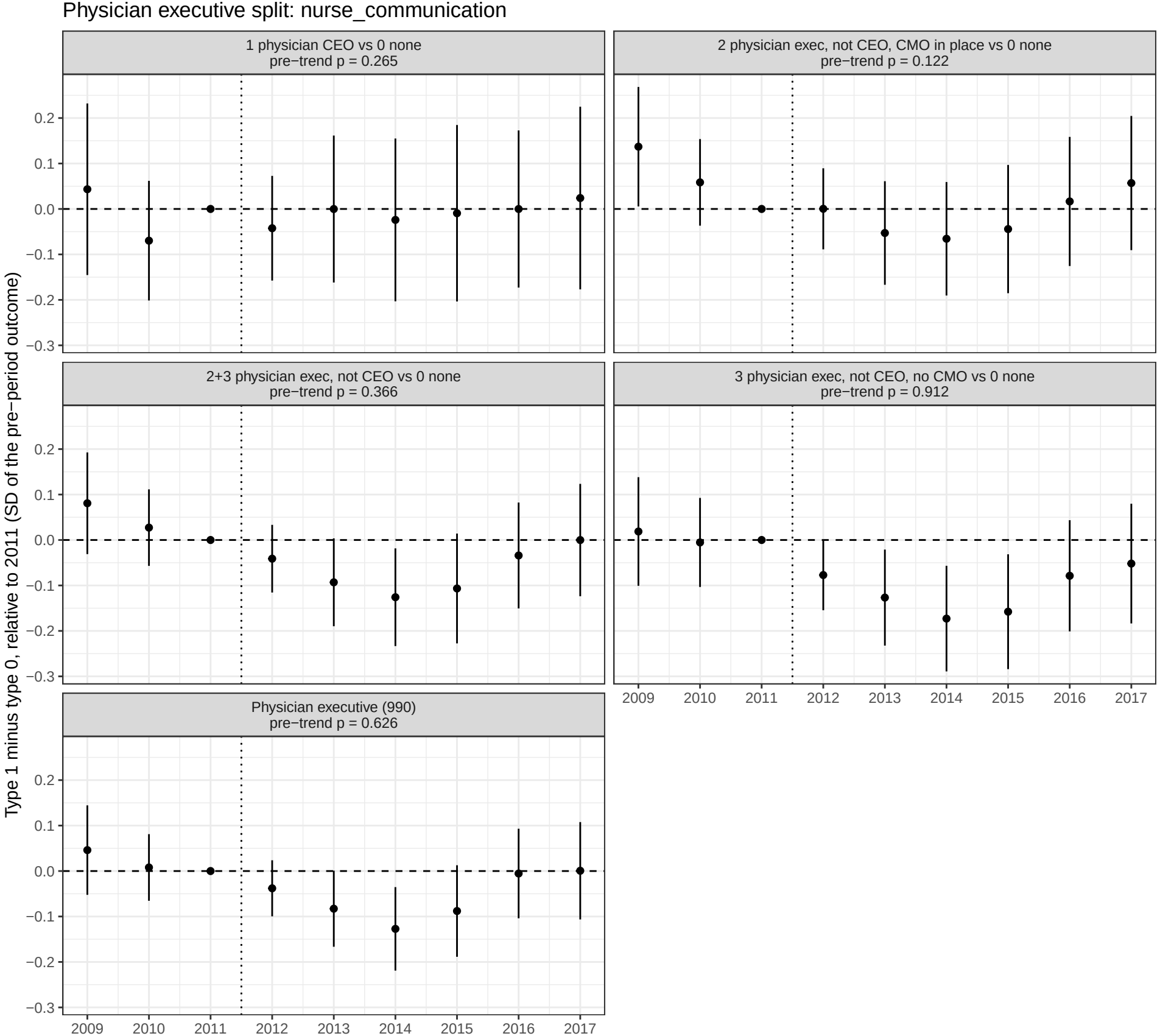

Physician executive split: staff_responsiveness
1 physician CEO vs 0 none
pre−trend p = 0.893
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.904
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.877
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.836
Physician executive (990)
pre−trend p = 0.855
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.0
−0.2
2009
2010
2011
2012
2013
2014
2015
2016
2017

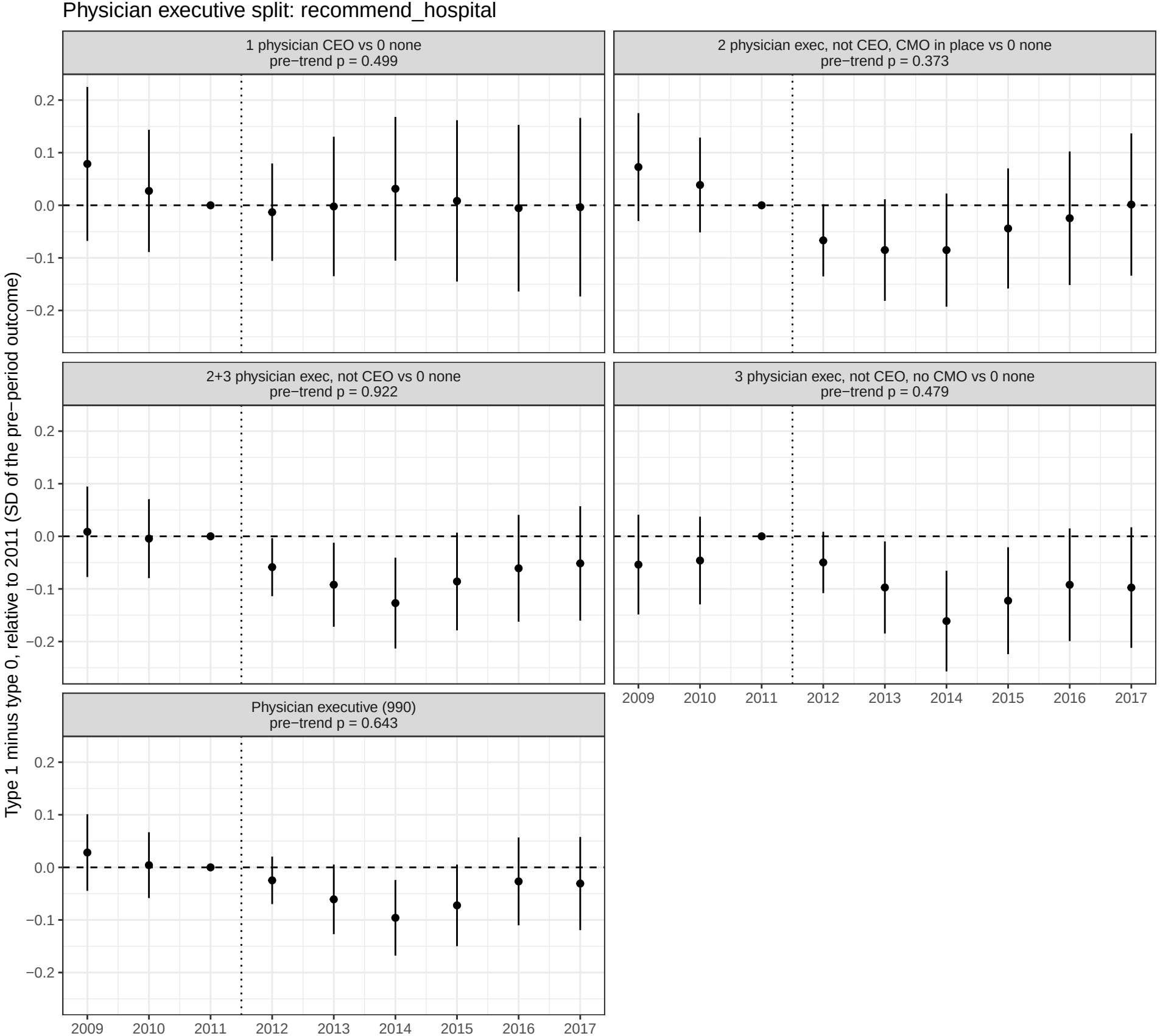

Physician executive split: medicine_communication
1 physician CEO vs 0 none
pre−trend p = 0.12
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.878
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.663
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.216
Physician executive (990)
pre−trend p = 0.527
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.0
−0.2
−0.4
−0.6
2009
2010
2011
2012
2013
2014
2015
2016
2017

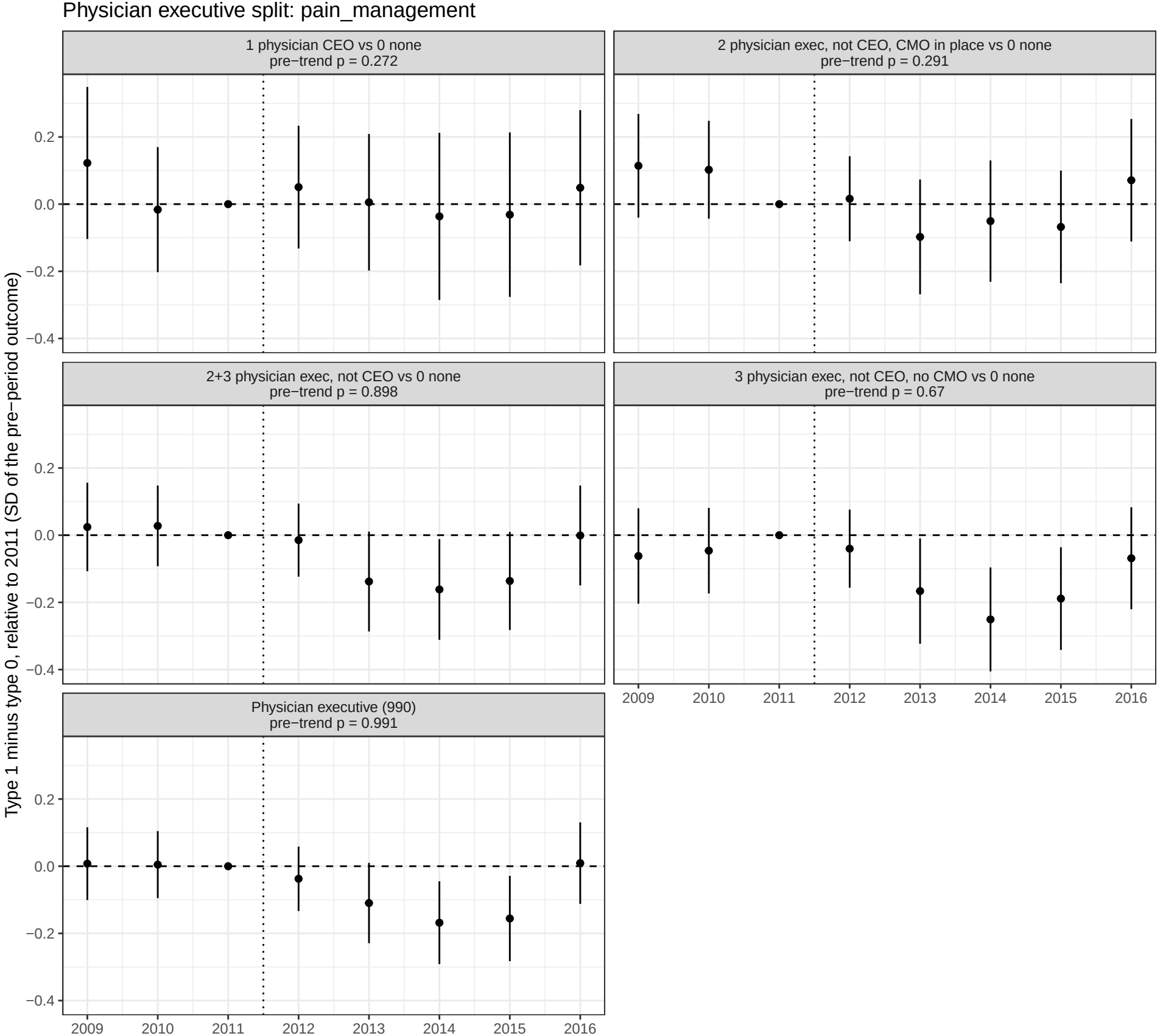

Physician executive split: doctor_communication
1 physician CEO vs 0 none
pre−trend p = 0.926
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.335
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.156
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.147
Physician executive (990)
pre−trend p = 0.593
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.1
0.0
−0.1
−0.2
2009
2010
2011
2012
2013
2014
2015
2016
2017

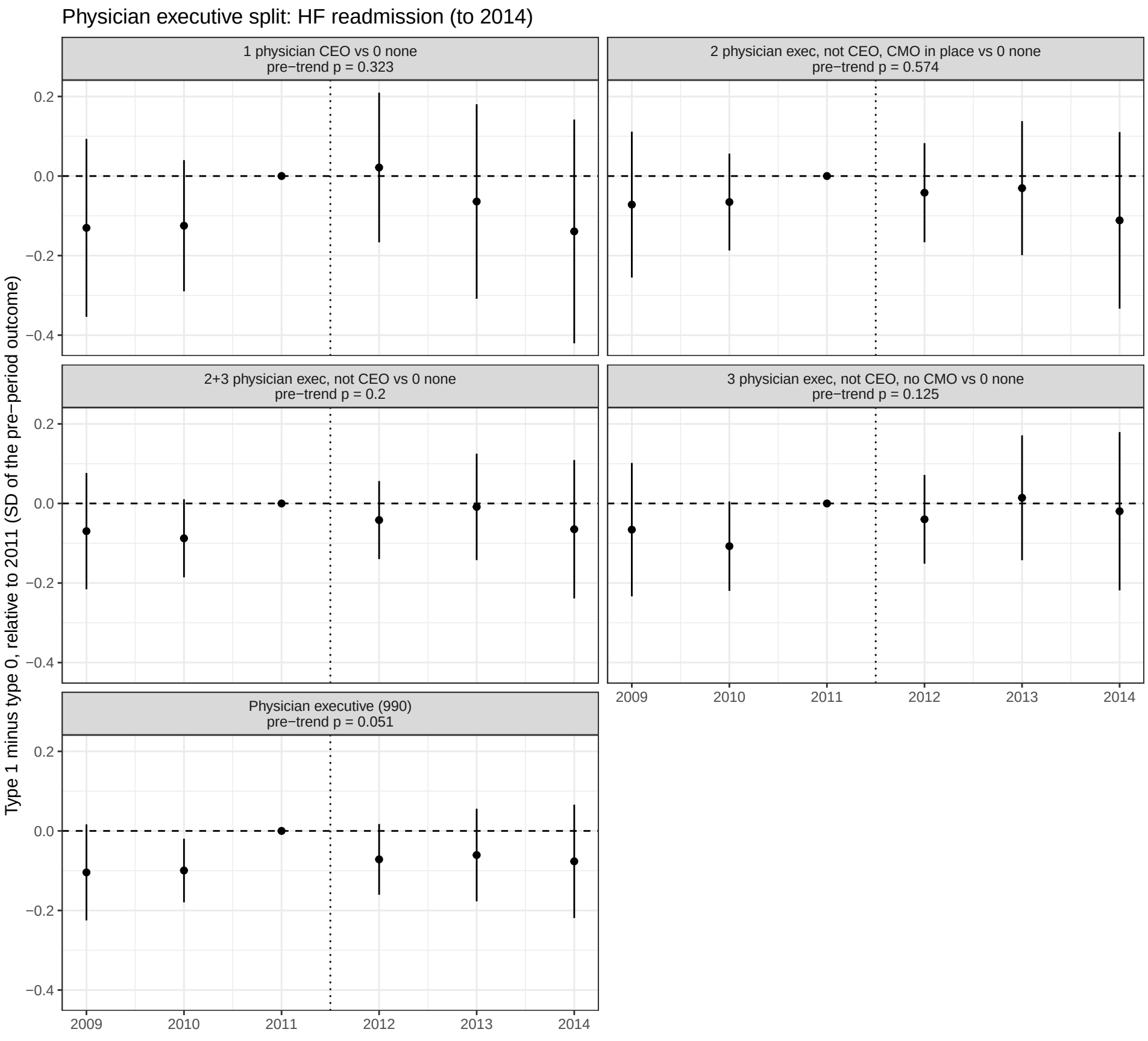

Physician executive split: cleanliness
1 physician CEO vs 0 none
pre−trend p = 0.847
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.05
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.116
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.481
Physician executive (990)
pre−trend p = 0.132
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.0
−0.2
−0.4
2009
2010
2011
2012
2013
2014
2015
2016
2017

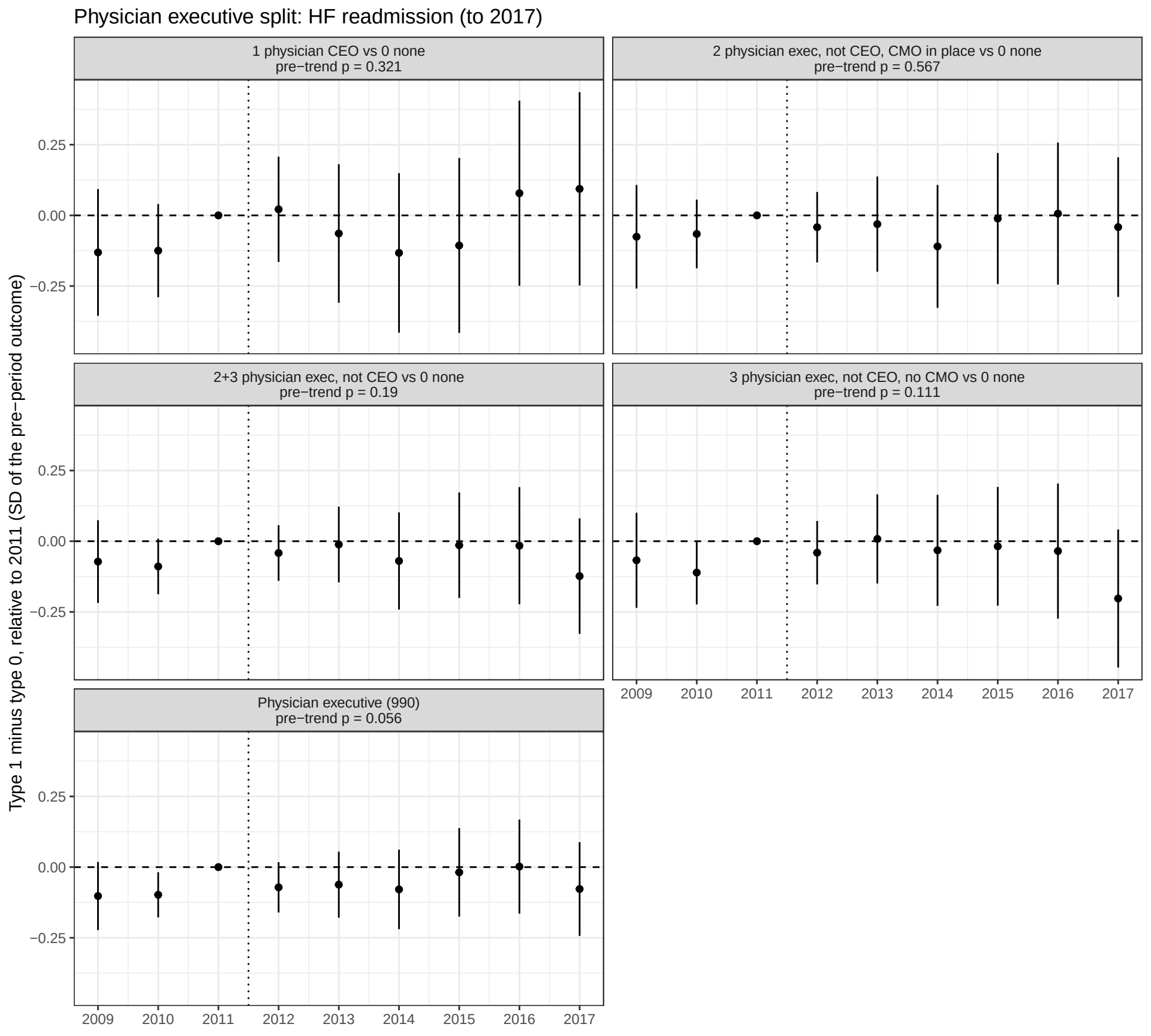

Physician executive split: quietness
1 physician CEO vs 0 none
pre−trend p = 0.112
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.043
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.228
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.887
Physician executive (990)
pre−trend p = 0.133
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.3
0.2
0.1
0.0
−0.1
−0.2
2009
2010
2011
2012
2013
2014
2015
2016
2017

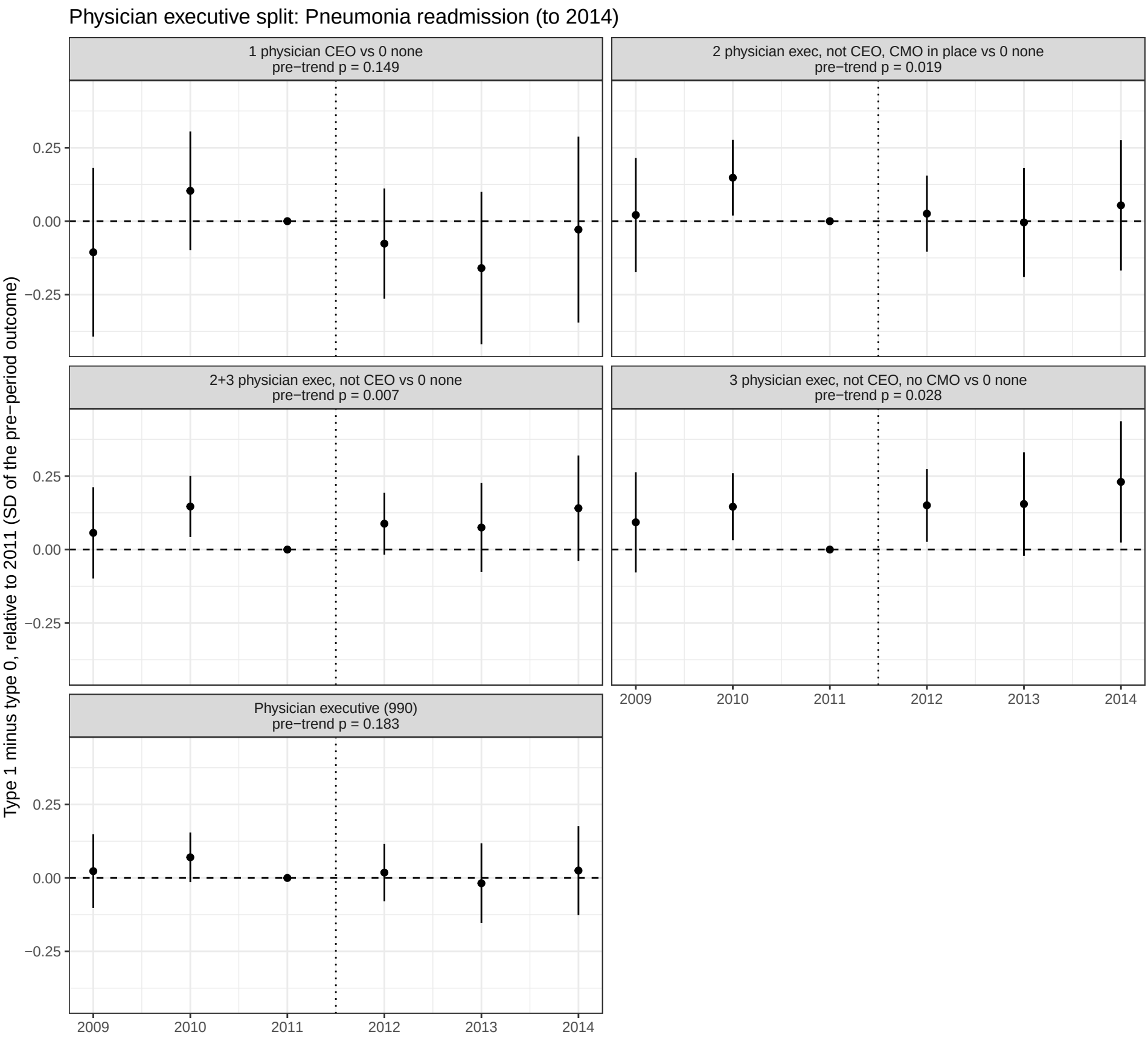

Physician executive split: nurse_communication
1 physician CEO vs 0 none
pre−trend p = 0.265
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.122
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.366
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.912
Physician executive (990)
pre−trend p = 0.626
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.1
0.0
−0.1
−0.2
−0.3
2009
2010
2011
2012
2013
2014
2015
2016
2017

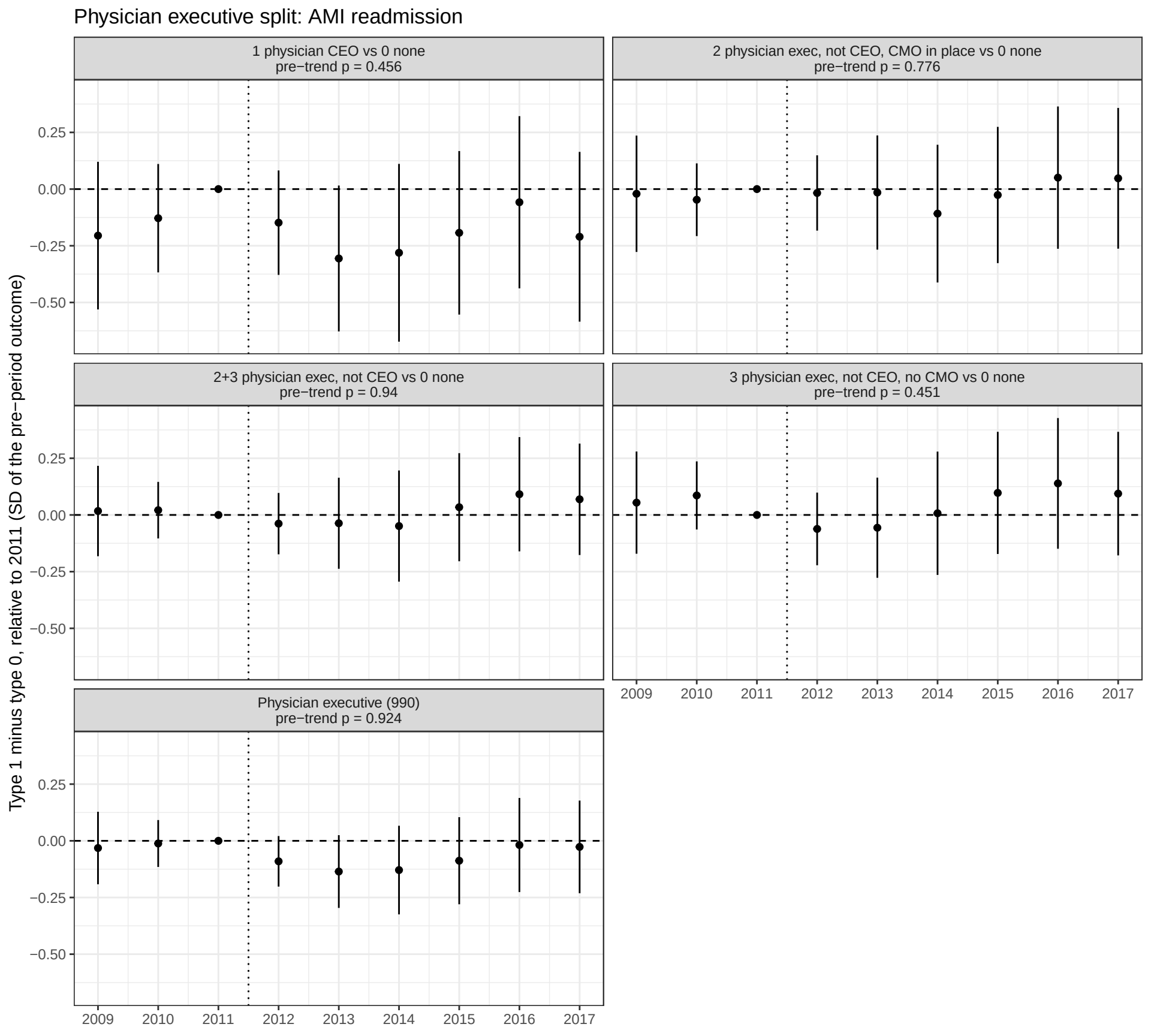

Physician executive split: recommend_hospital
1 physician CEO vs 0 none
pre−trend p = 0.499
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.373
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.922
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.479
Physician executive (990)
pre−trend p = 0.643
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.1
0.0
−0.1
−0.2
2009
2010
2011
2012
2013
2014
2015
2016
2017

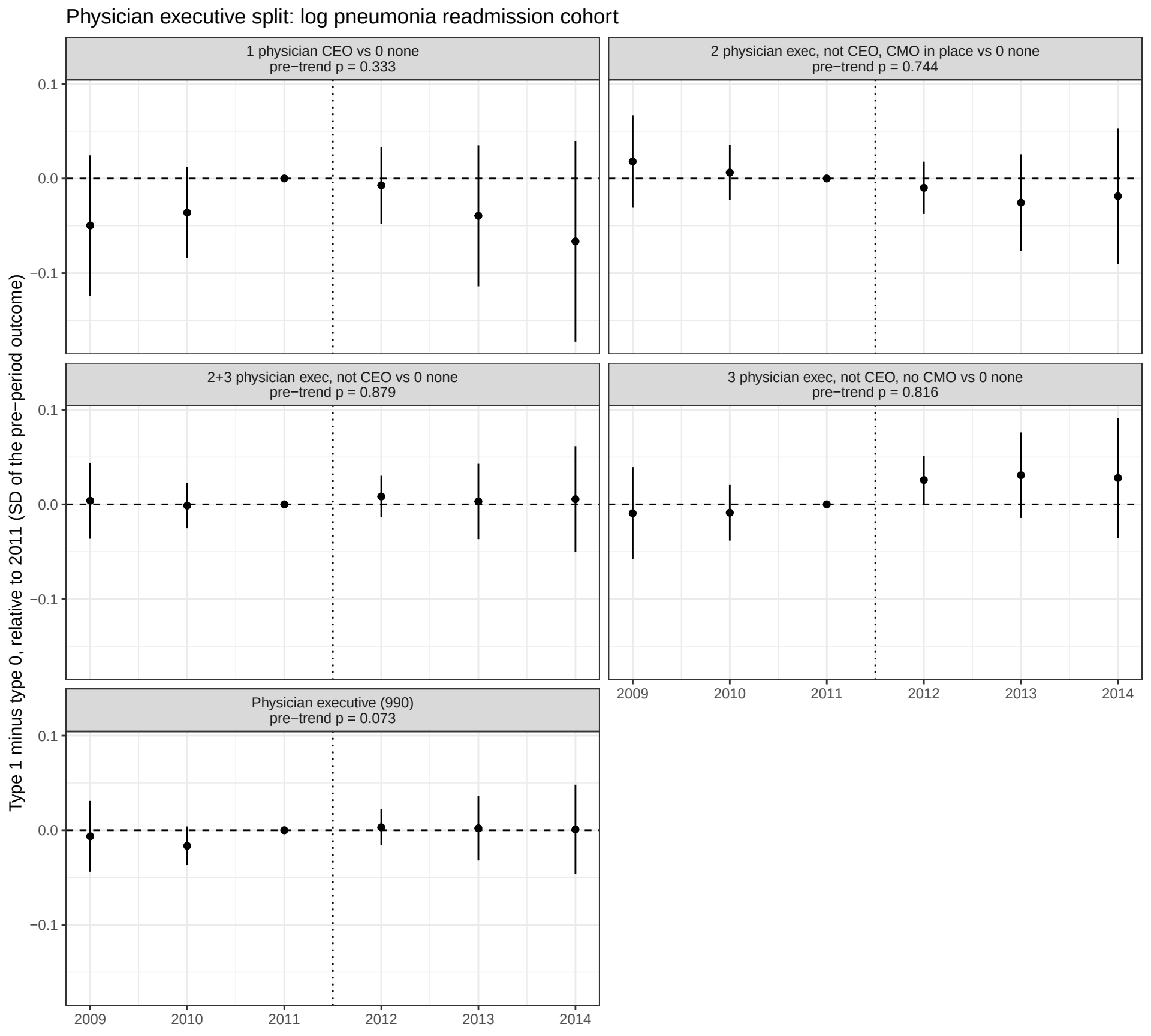

Physician executive split: pain_management
1 physician CEO vs 0 none
pre−trend p = 0.272
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.291
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.898
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.67
Physician executive (990)
pre−trend p = 0.991
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.0
−0.2
−0.4
2009
2010
2011
2012
2013
2014
2015
2016

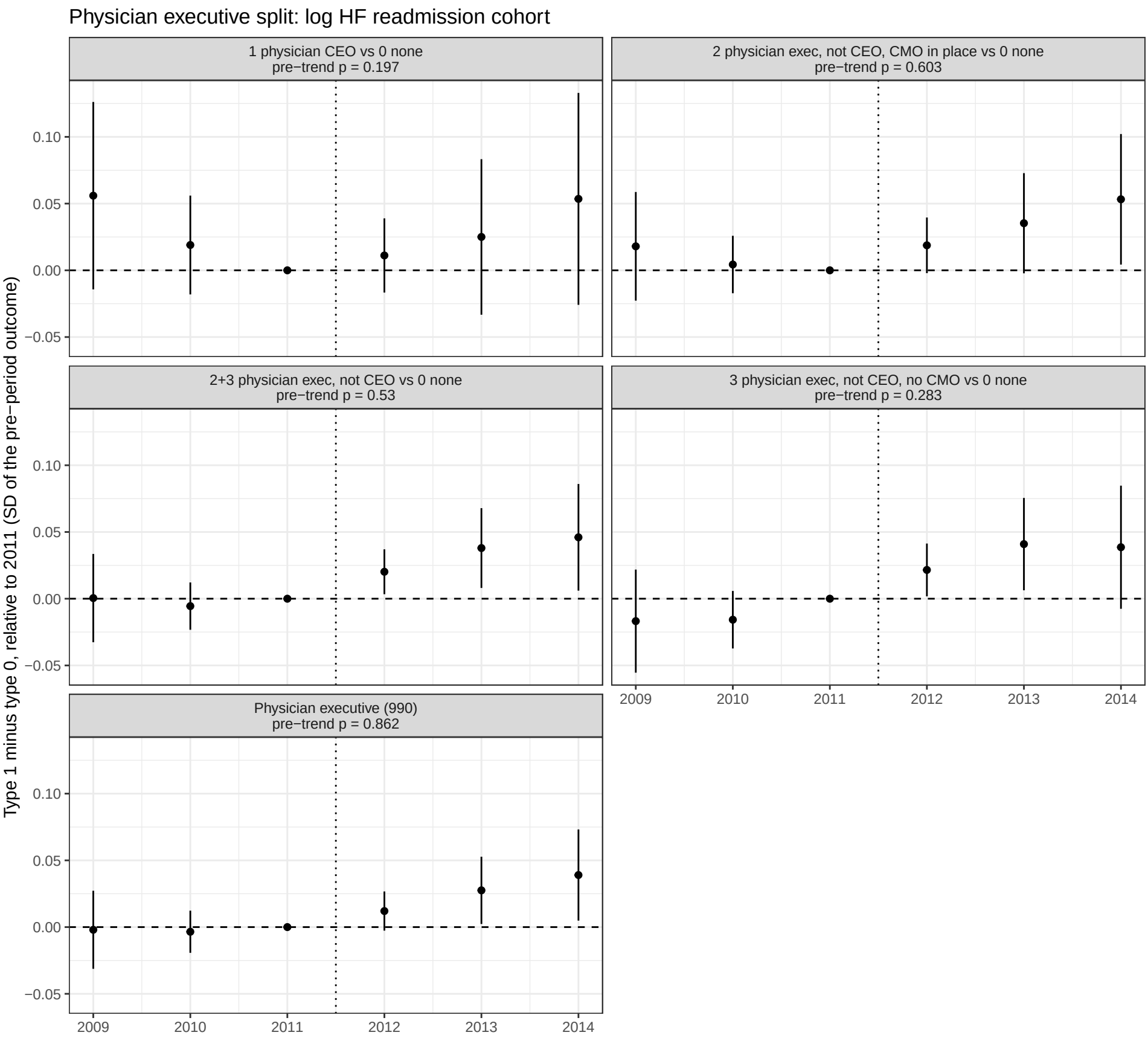

Physician executive split: HF readmission (to 2014)
1 physician CEO vs 0 none
pre−trend p = 0.323
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.574
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.2
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.125
Physician executive (990)
pre−trend p = 0.051
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.0
−0.2
−0.4
2009
2010
2011
2012
2013
2014

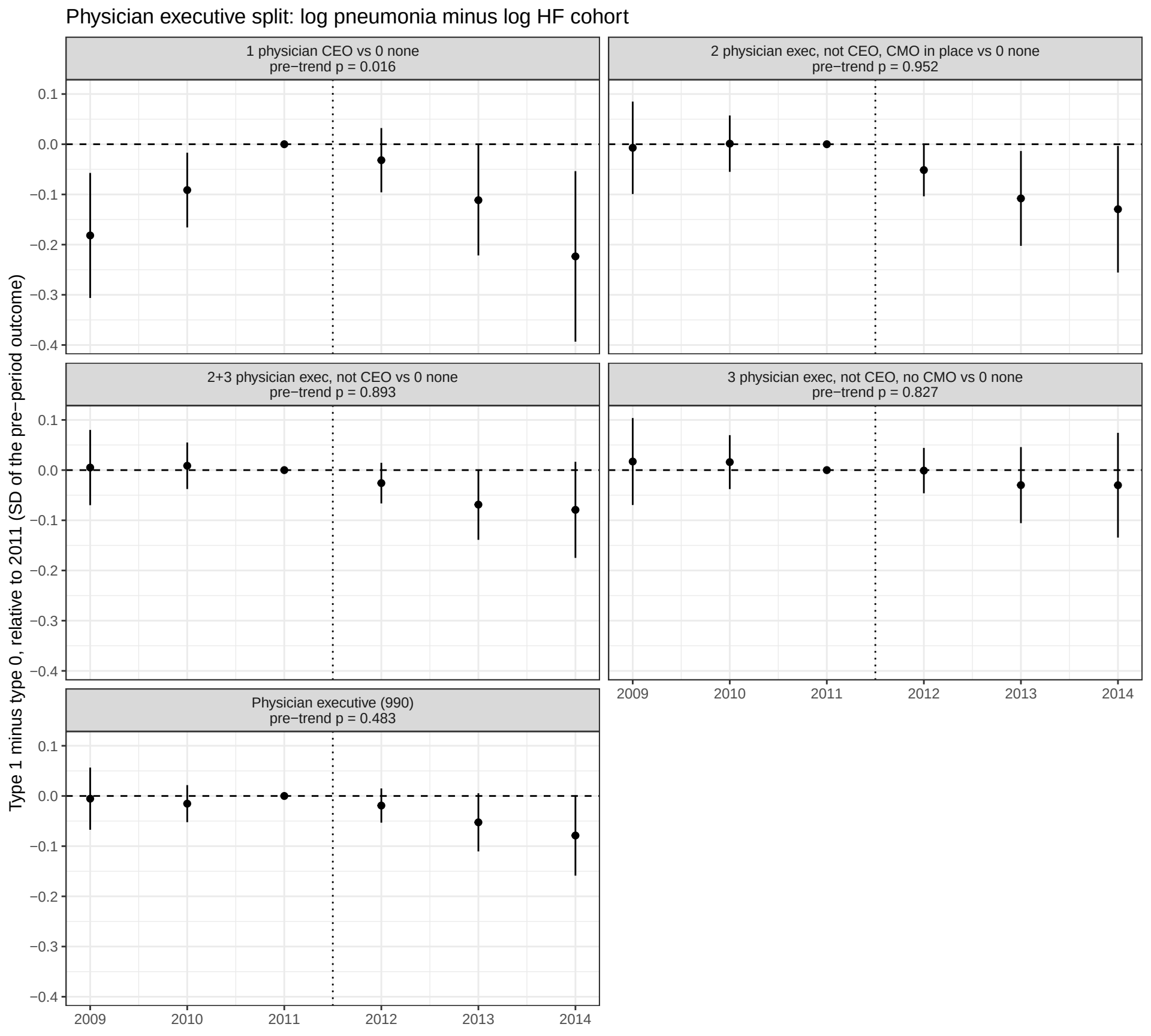

Physician executive split: HF readmission (to 2017)
1 physician CEO vs 0 none
pre−trend p = 0.321
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.567
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.19
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.111
Physician executive (990)
pre−trend p = 0.056
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.25
0.00
−0.25
2009
2010
2011
2012
2013
2014
2015
2016
2017

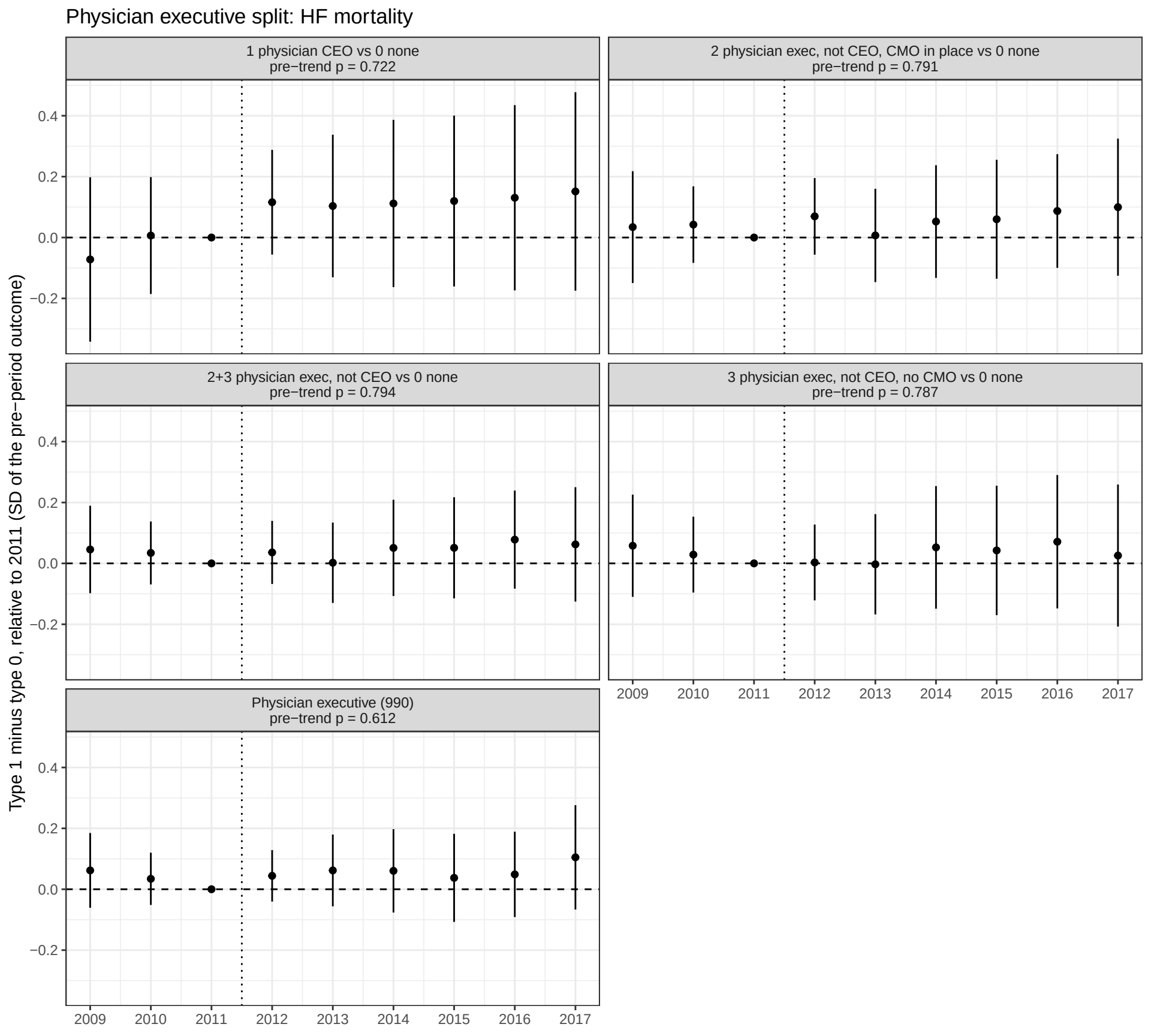

Physician executive split: Pneumonia readmission (to 2014)
1 physician CEO vs 0 none
pre−trend p = 0.149
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.019
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.007
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.028
Physician executive (990)
pre−trend p = 0.183
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.25
0.00
−0.25
2009
2010
2011
2012
2013
2014

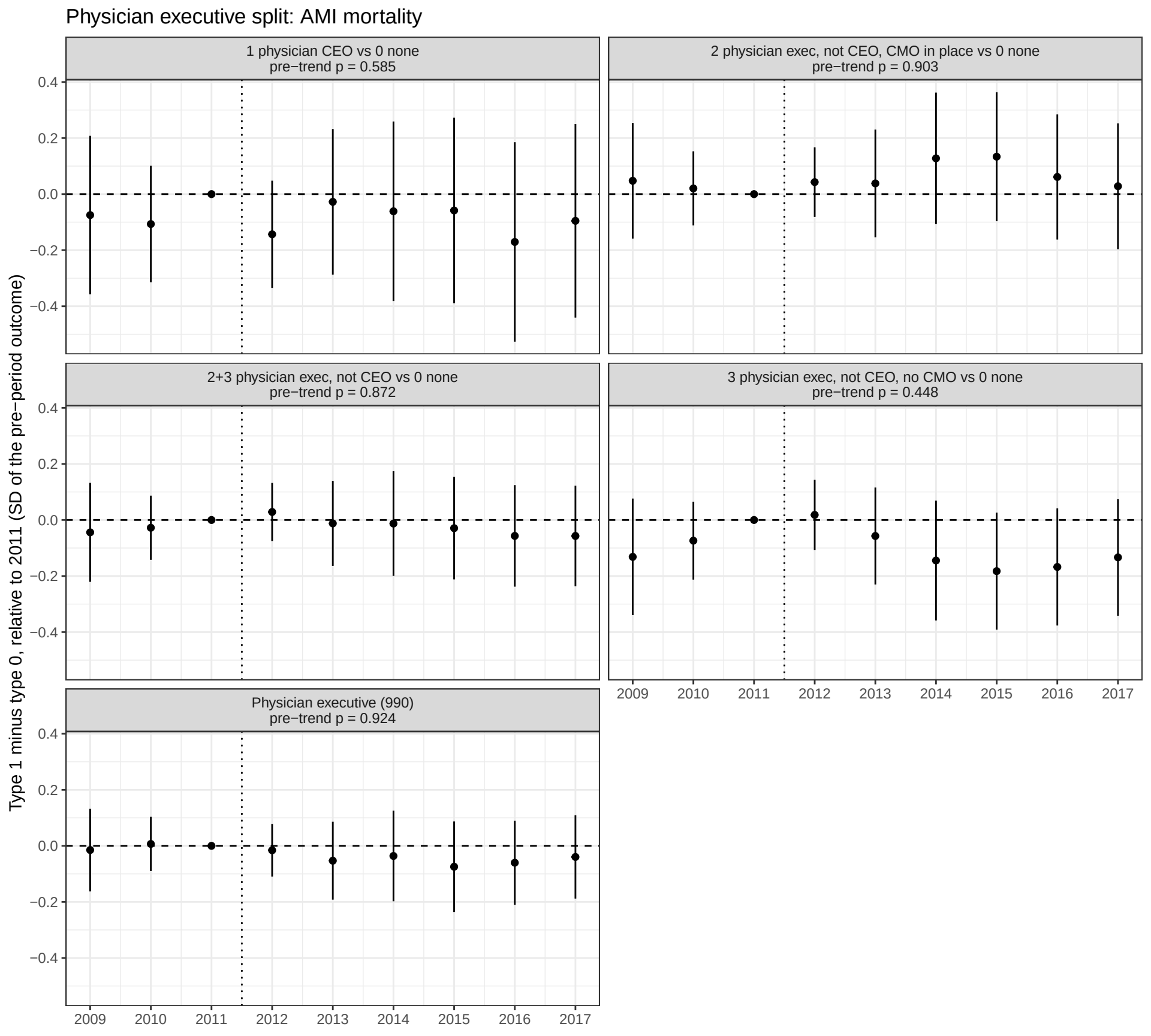

Physician executive split: AMI readmission
1 physician CEO vs 0 none
pre−trend p = 0.456
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.776
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.94
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.451
Physician executive (990)
pre−trend p = 0.924
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.25
0.00
−0.25
−0.50
2009
2010
2011
2012
2013
2014
2015
2016
2017

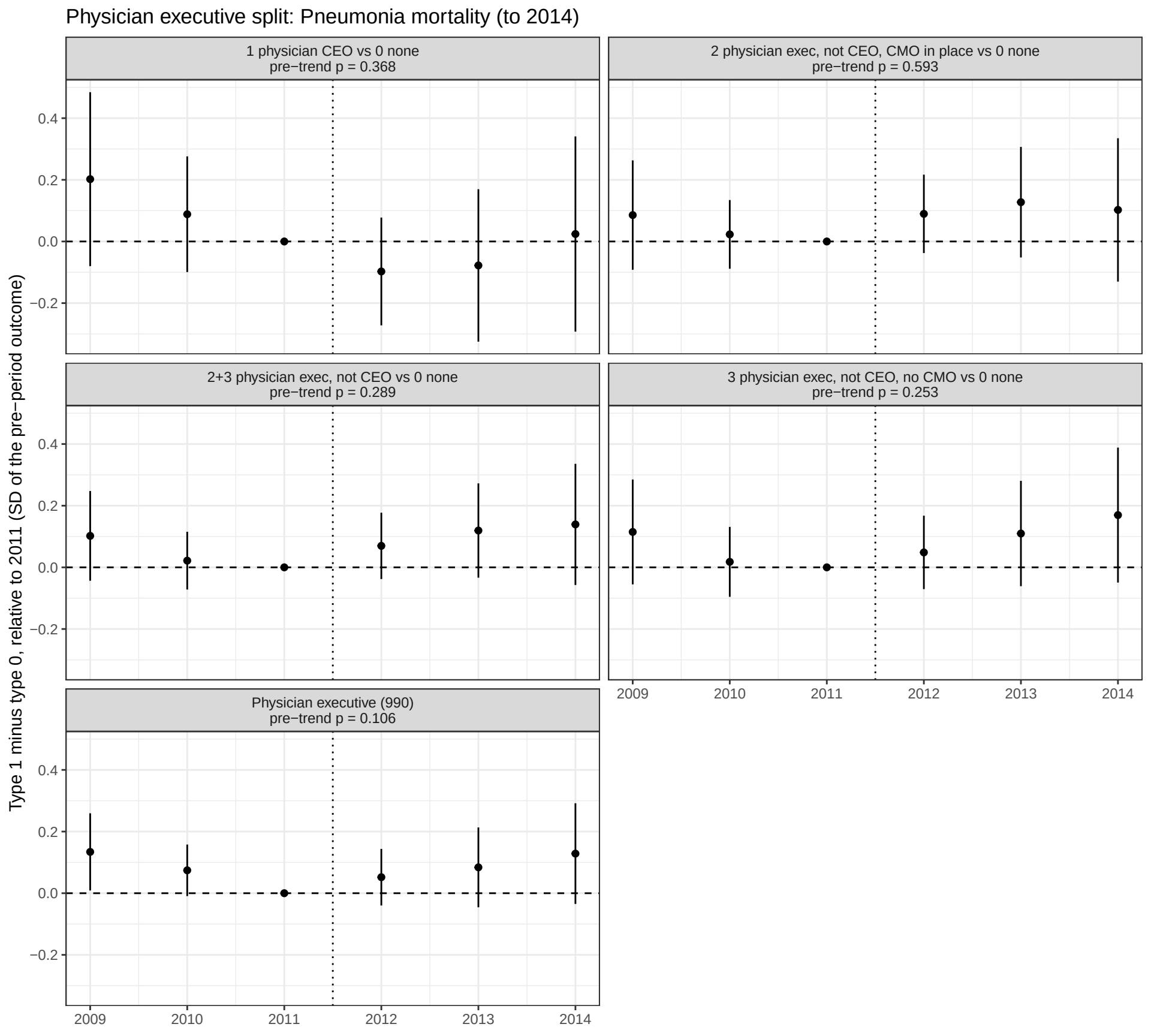

Physician executive split: log pneumonia readmission cohort
1 physician CEO vs 0 none
pre−trend p = 0.333
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.744
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.879
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.816
Physician executive (990)
pre−trend p = 0.073
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.1
0.0
−0.1
2009
2010
2011
2012
2013
2014

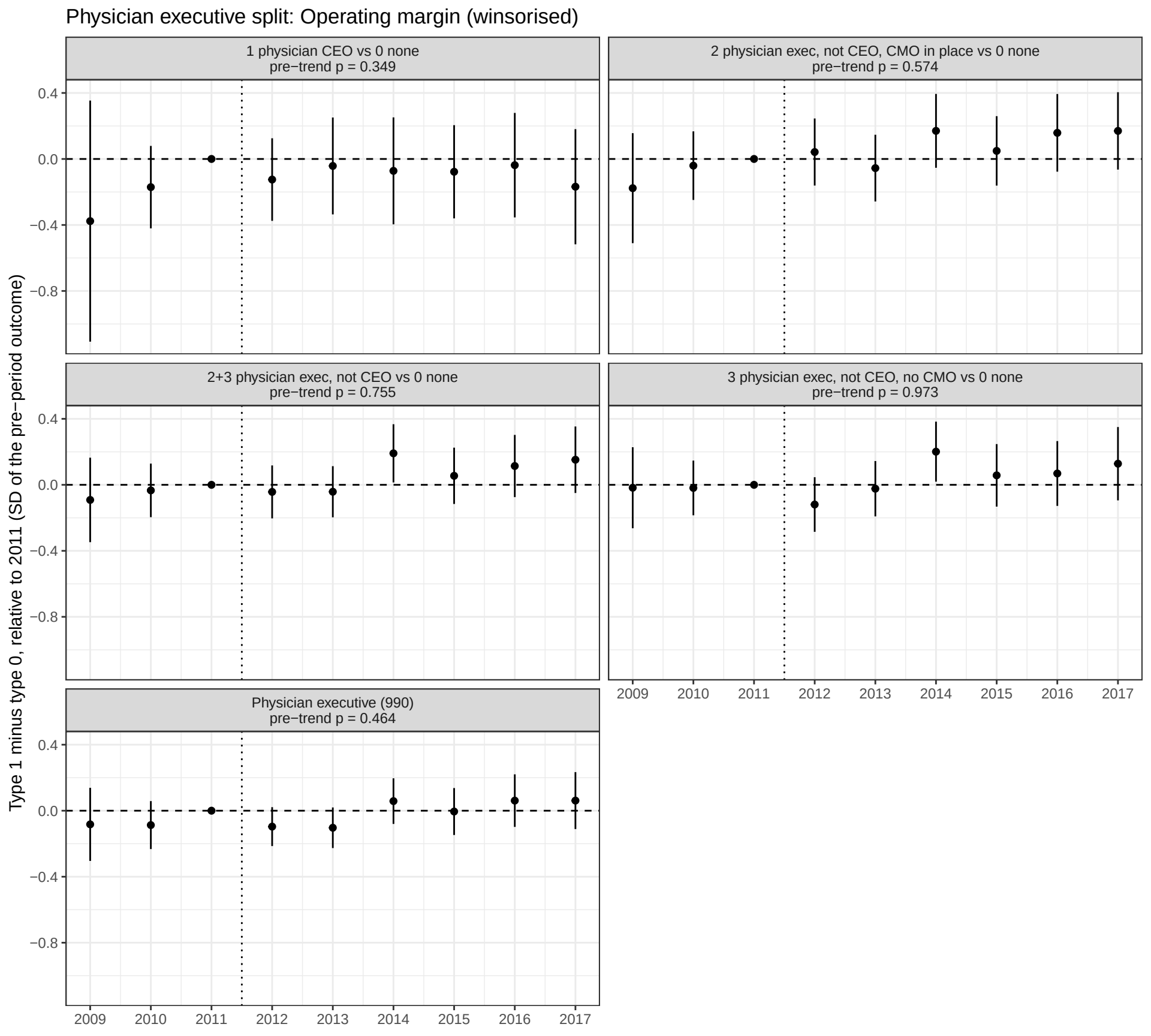

Physician executive split: log HF readmission cohort
1 physician CEO vs 0 none
pre−trend p = 0.197
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.603
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.53
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.283
Physician executive (990)
pre−trend p = 0.862
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.10
0.05
0.00
−0.05
2009
2010
2011
2012
2013
2014

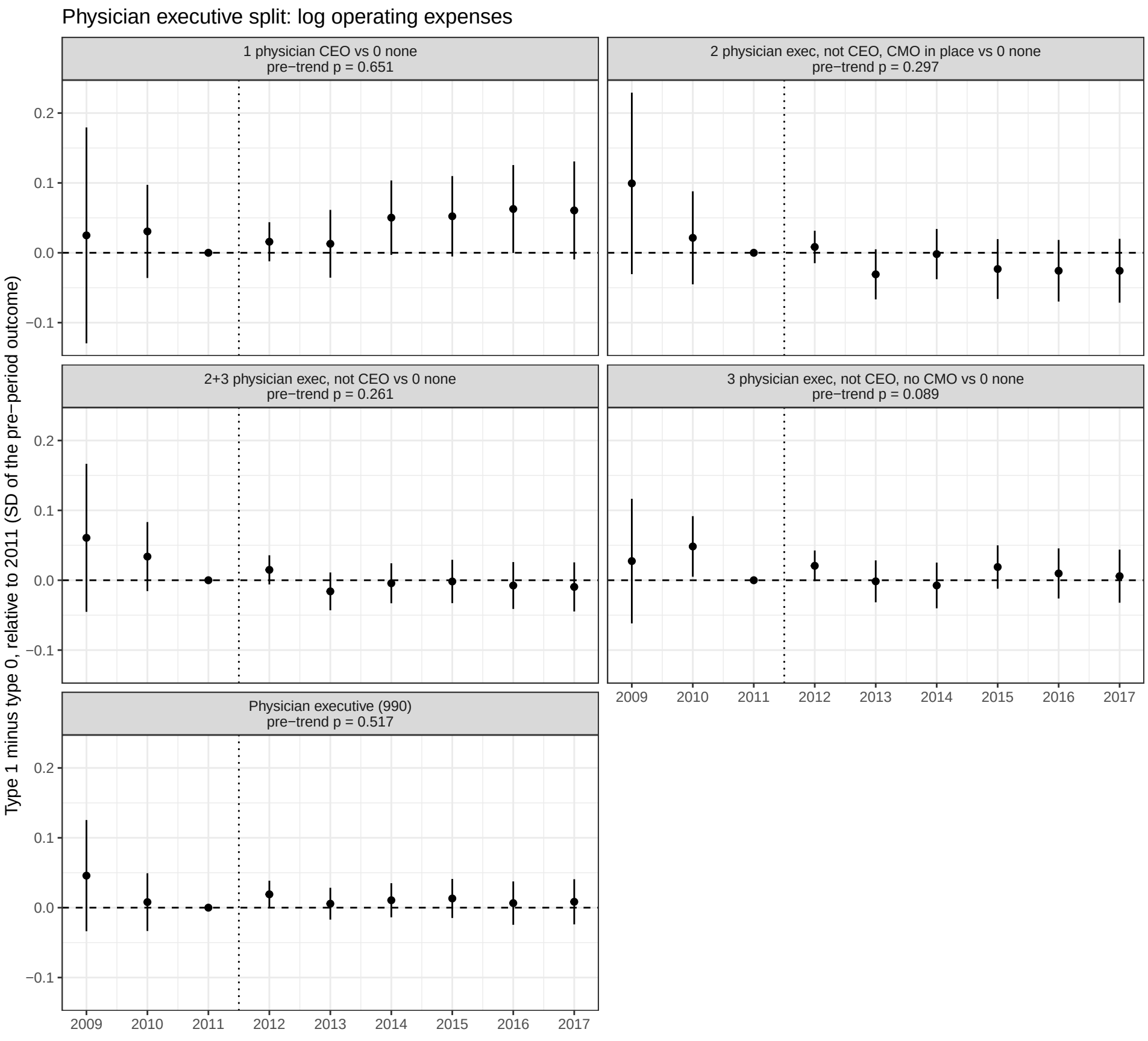

Physician executive split: log pneumonia minus log HF cohort
1 physician CEO vs 0 none
pre−trend p = 0.016
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.952
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.893
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.827
Physician executive (990)
pre−trend p = 0.483
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.1
0.0
−0.1
−0.2
−0.3
−0.4
2009
2010
2011
2012
2013
2014

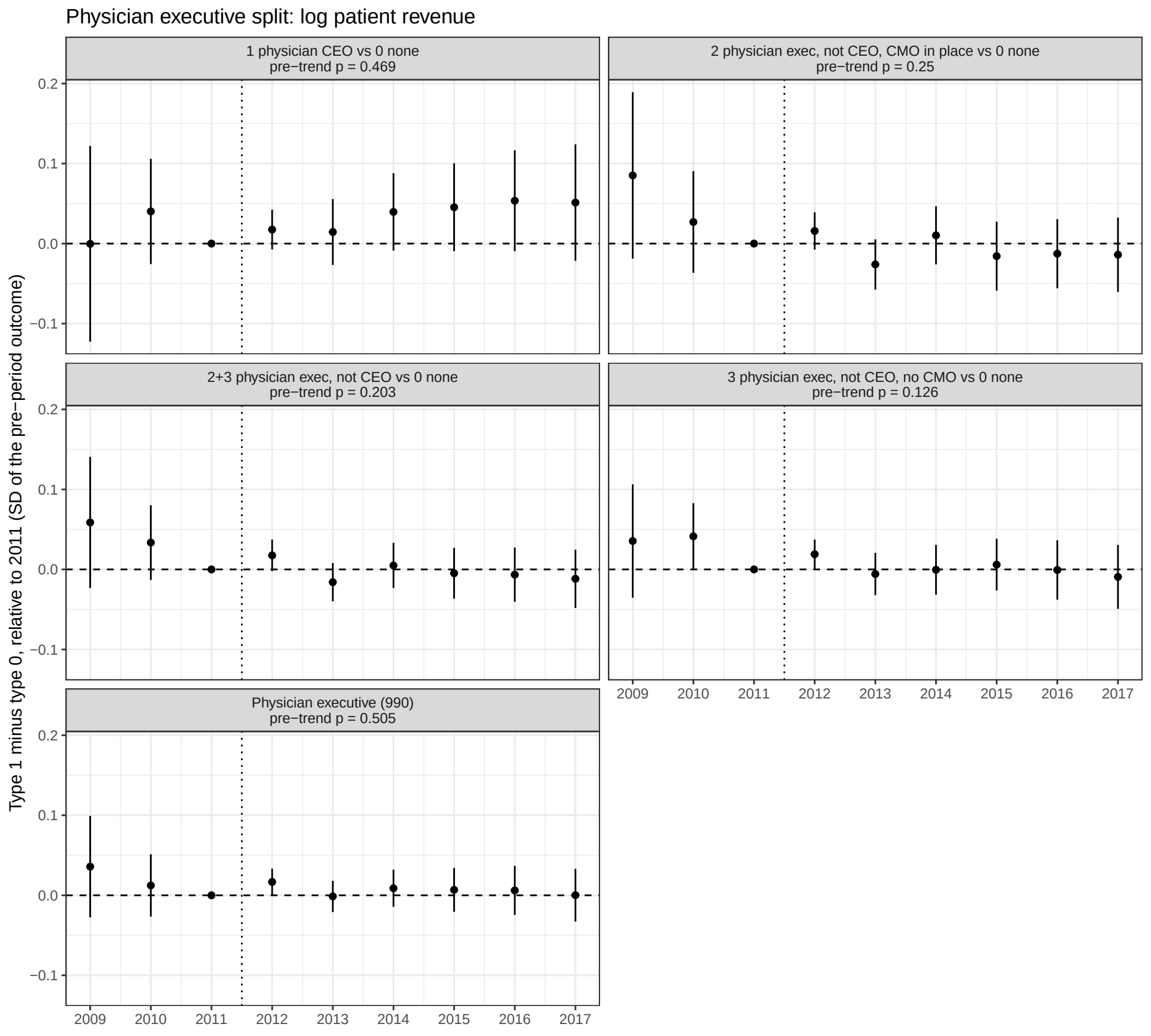

Physician executive split: HF mortality
1 physician CEO vs 0 none
pre−trend p = 0.722
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.791
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.794
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.787
Physician executive (990)
pre−trend p = 0.612
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.4
0.2
0.0
−0.2
2009
2010
2011
2012
2013
2014
2015
2016
2017

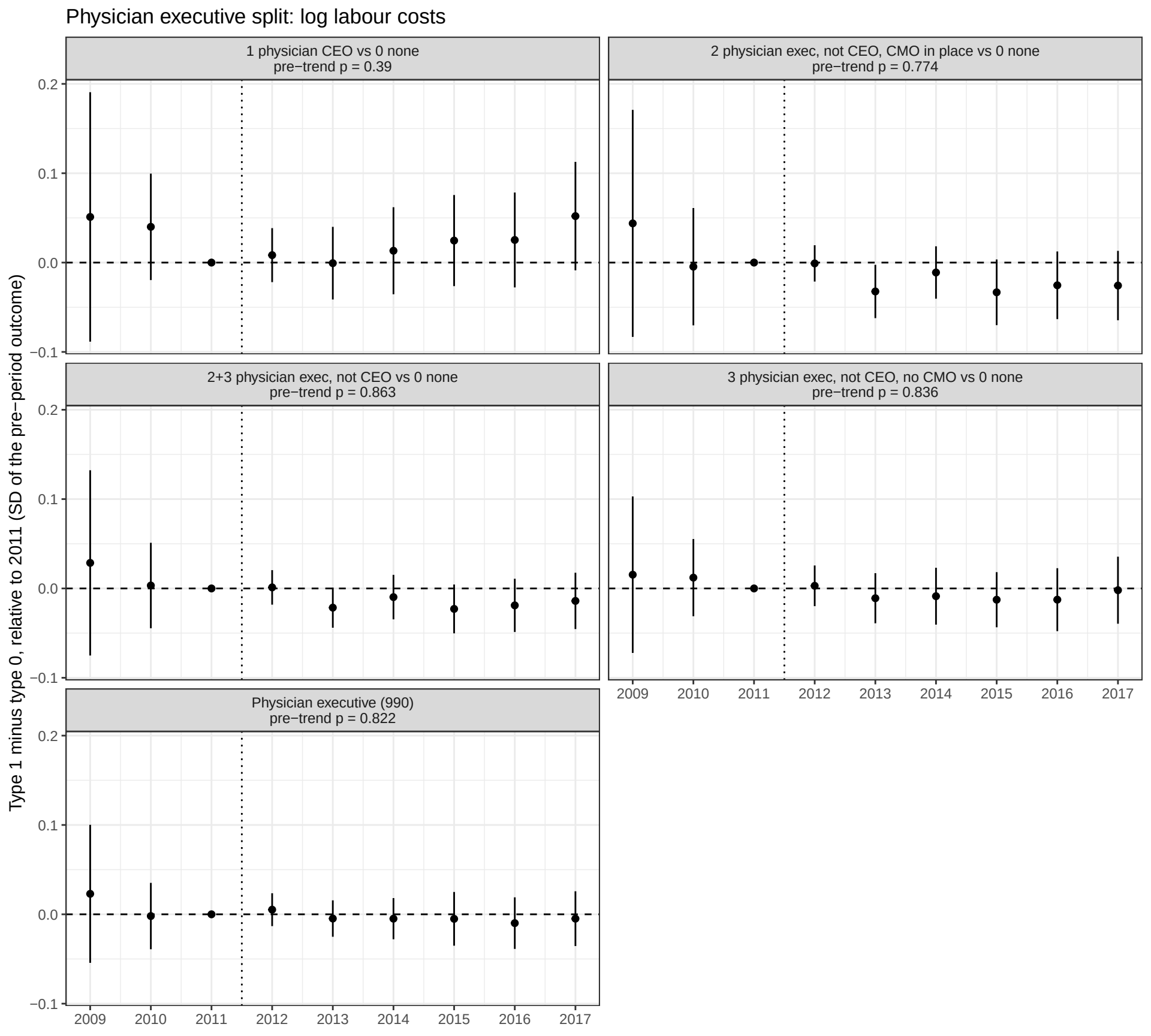

Physician executive split: AMI mortality
1 physician CEO vs 0 none
pre−trend p = 0.585
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.903
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.872
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.448
Physician executive (990)
pre−trend p = 0.924
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.4
0.2
0.0
−0.2
−0.4
2009
2010
2011
2012
2013
2014
2015
2016
2017

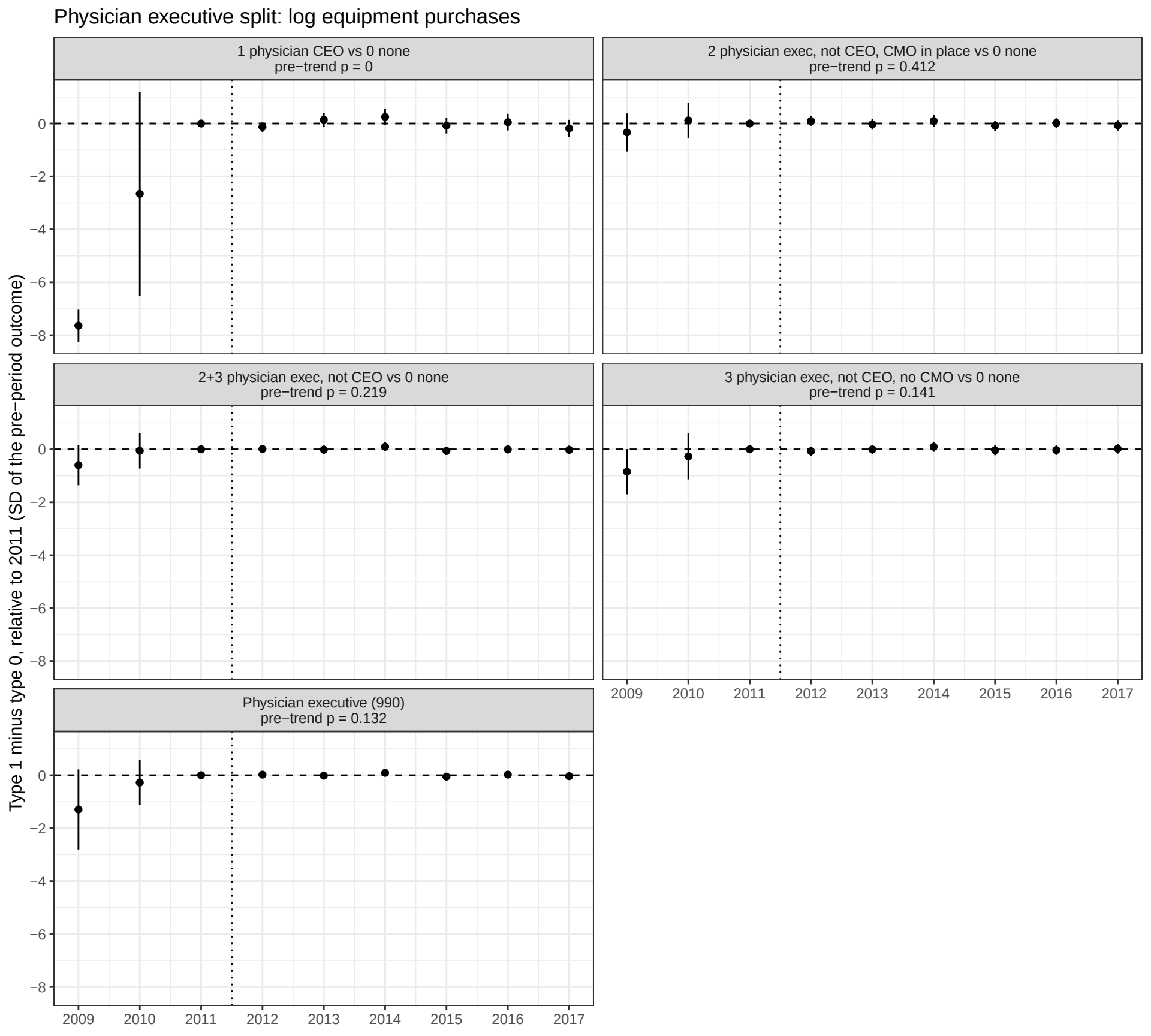

Physician executive split: Pneumonia mortality (to 2014)
1 physician CEO vs 0 none
pre−trend p = 0.368
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.593
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.289
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.253
Physician executive (990)
pre−trend p = 0.106
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.4
0.2
0.0
−0.2
2009
2010
2011
2012
2013
2014

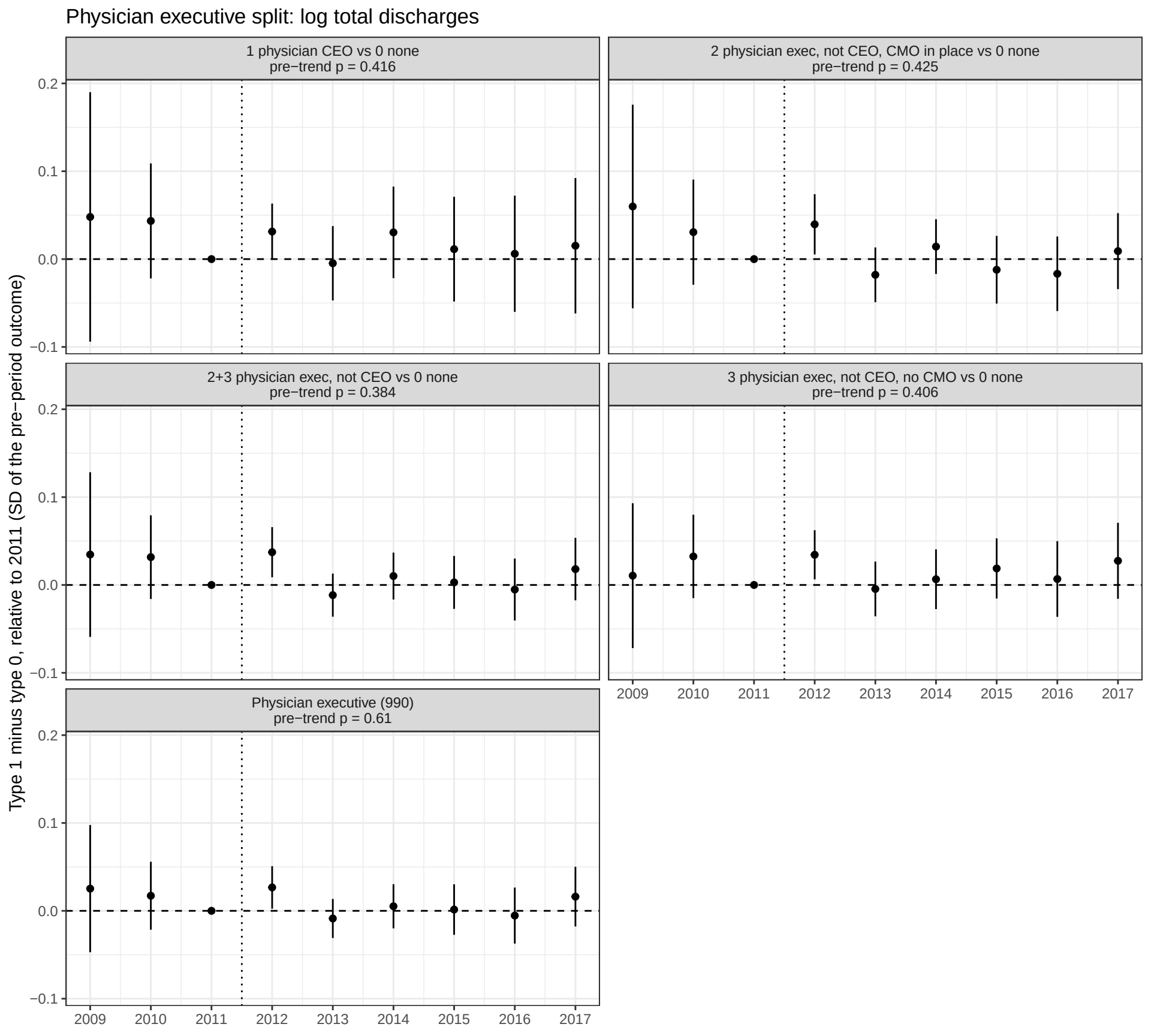

Physician executive split: Operating margin (winsorised)
1 physician CEO vs 0 none
pre−trend p = 0.349
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.574
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.755
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.973
Physician executive (990)
pre−trend p = 0.464
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.4
0.0
−0.4
−0.8
2009
2010
2011
2012
2013
2014
2015
2016
2017

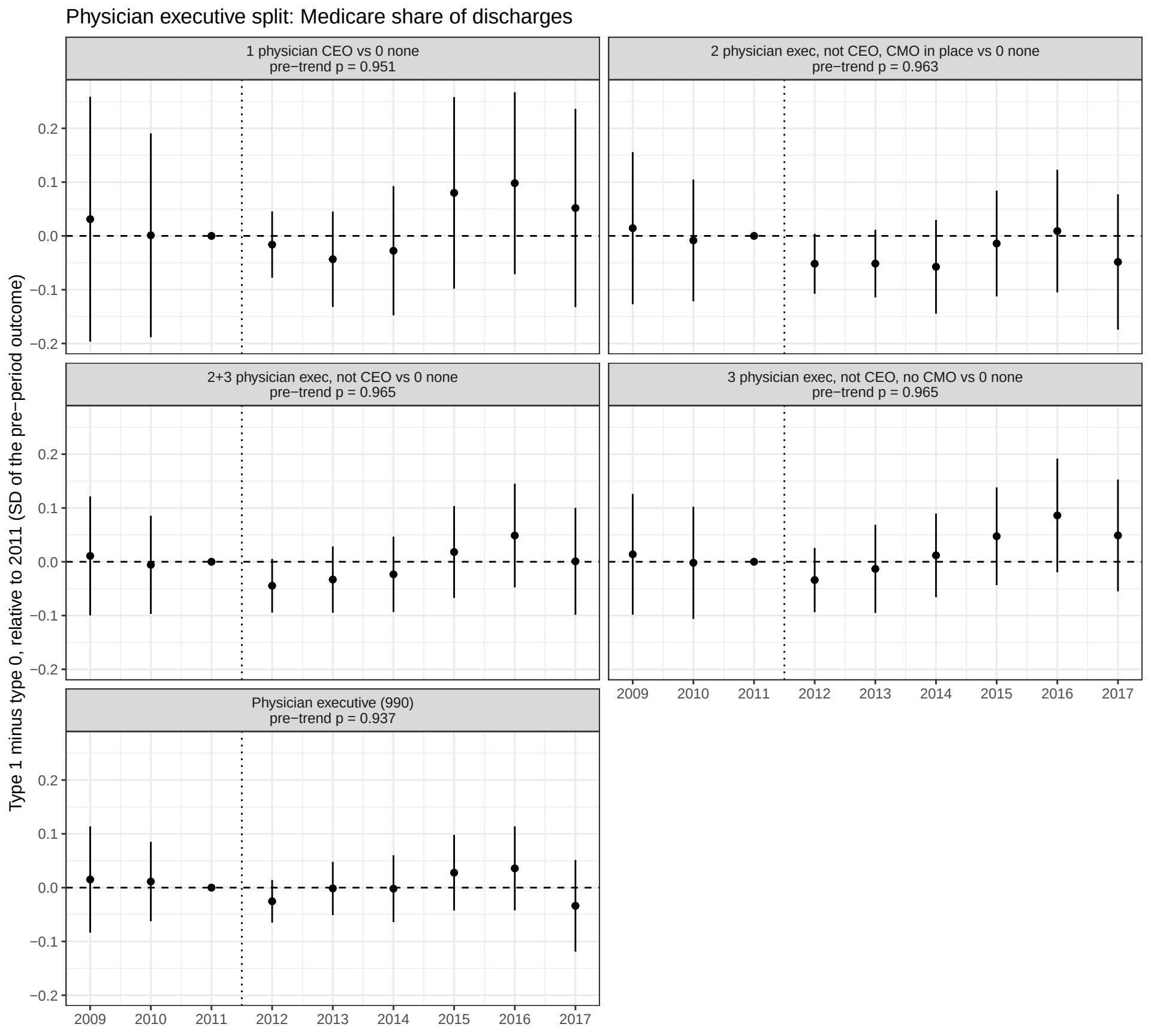

Physician executive split: log operating expenses
1 physician CEO vs 0 none
pre−trend p = 0.651
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.297
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.261
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.089
Physician executive (990)
pre−trend p = 0.517
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.1
0.0
−0.1
2009
2010
2011
2012
2013
2014
2015
2016
2017

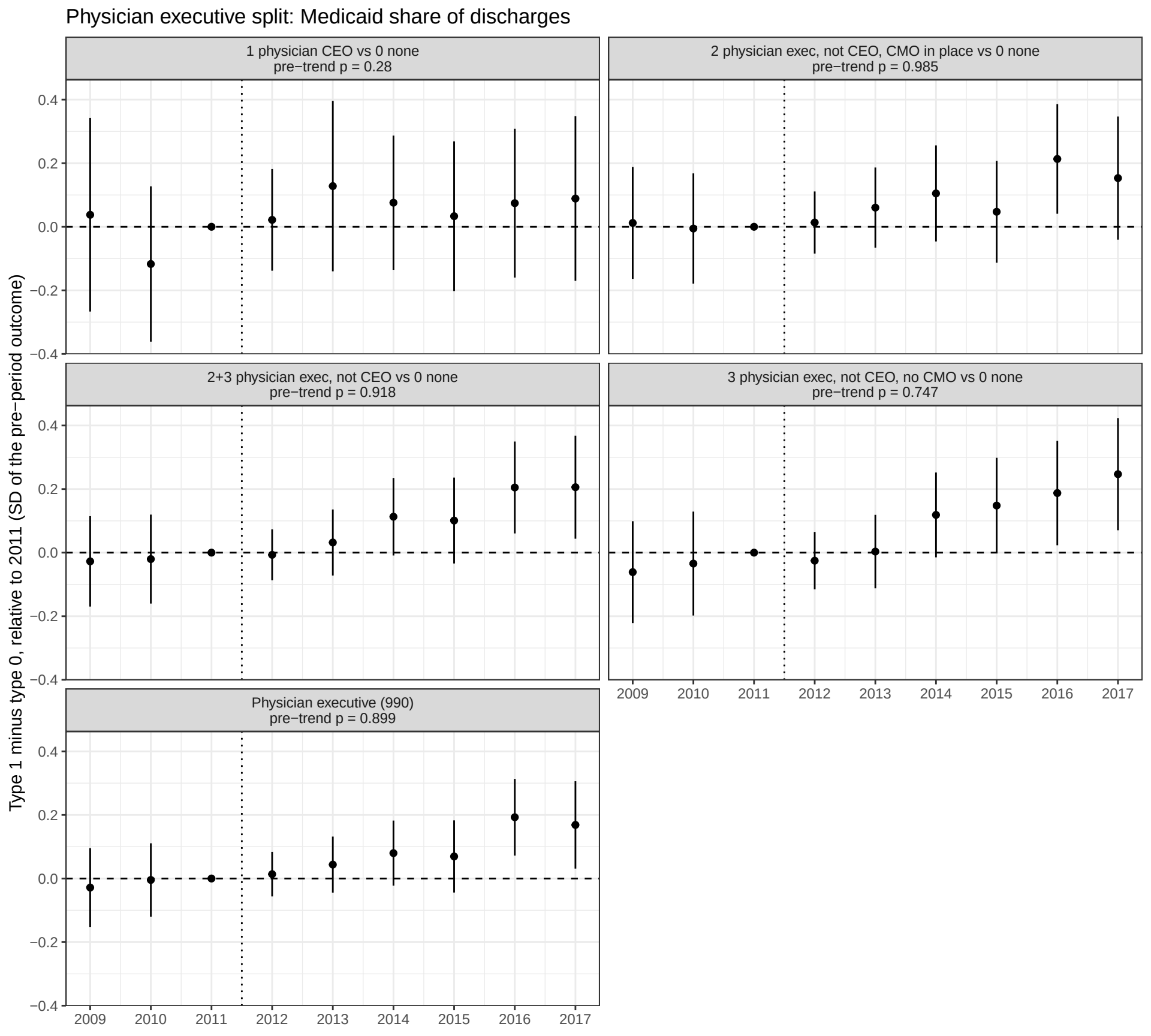

Physician executive split: log patient revenue
1 physician CEO vs 0 none
pre−trend p = 0.469
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.25
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.203
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.126
Physician executive (990)
pre−trend p = 0.505
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.1
0.0
−0.1
2009
2010
2011
2012
2013
2014
2015
2016
2017

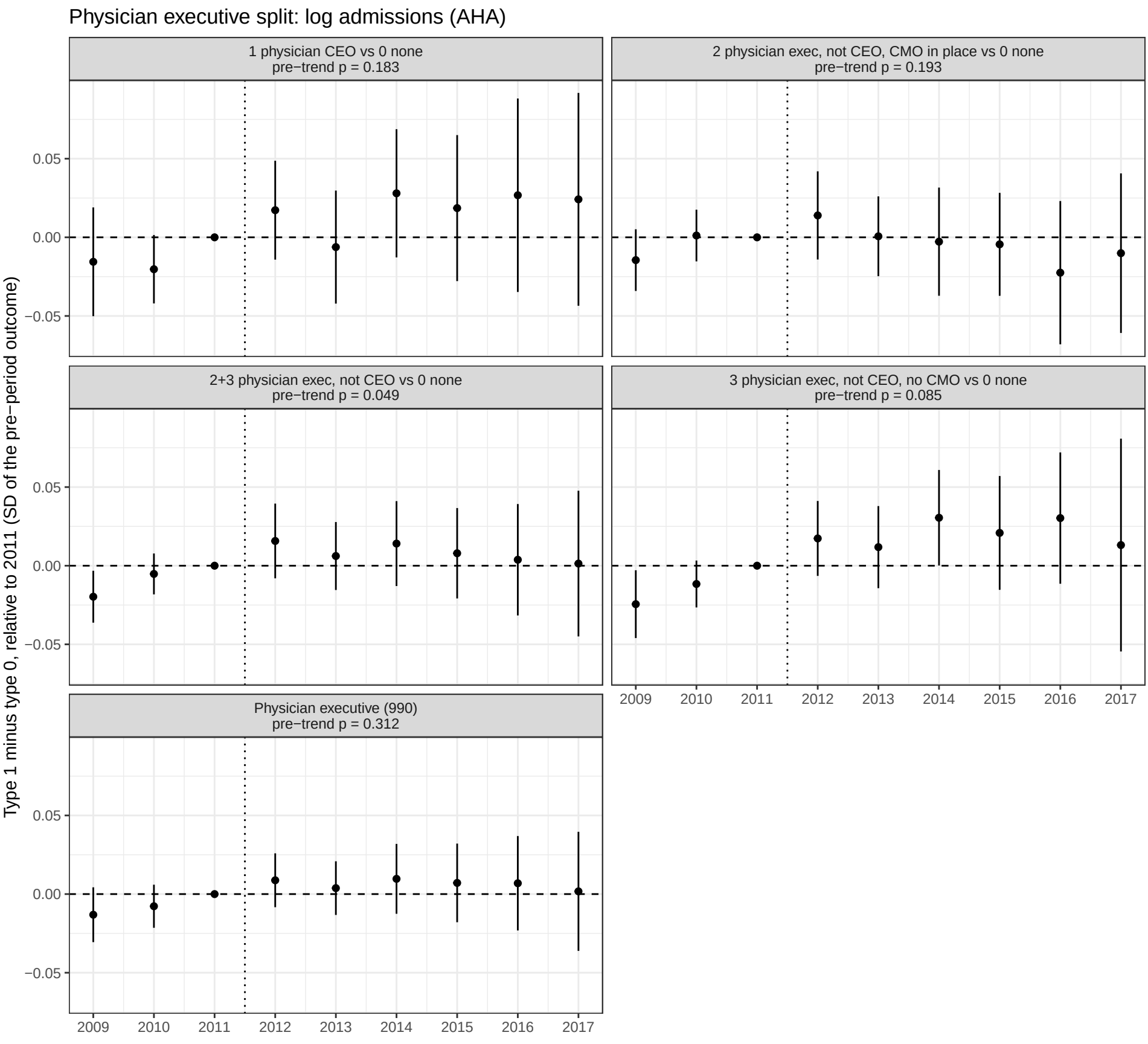

Physician executive split: log labour costs
1 physician CEO vs 0 none
pre−trend p = 0.39
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.774
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.863
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.836
Physician executive (990)
pre−trend p = 0.822
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.1
0.0
−0.1
2009
2010
2011
2012
2013
2014
2015
2016
2017

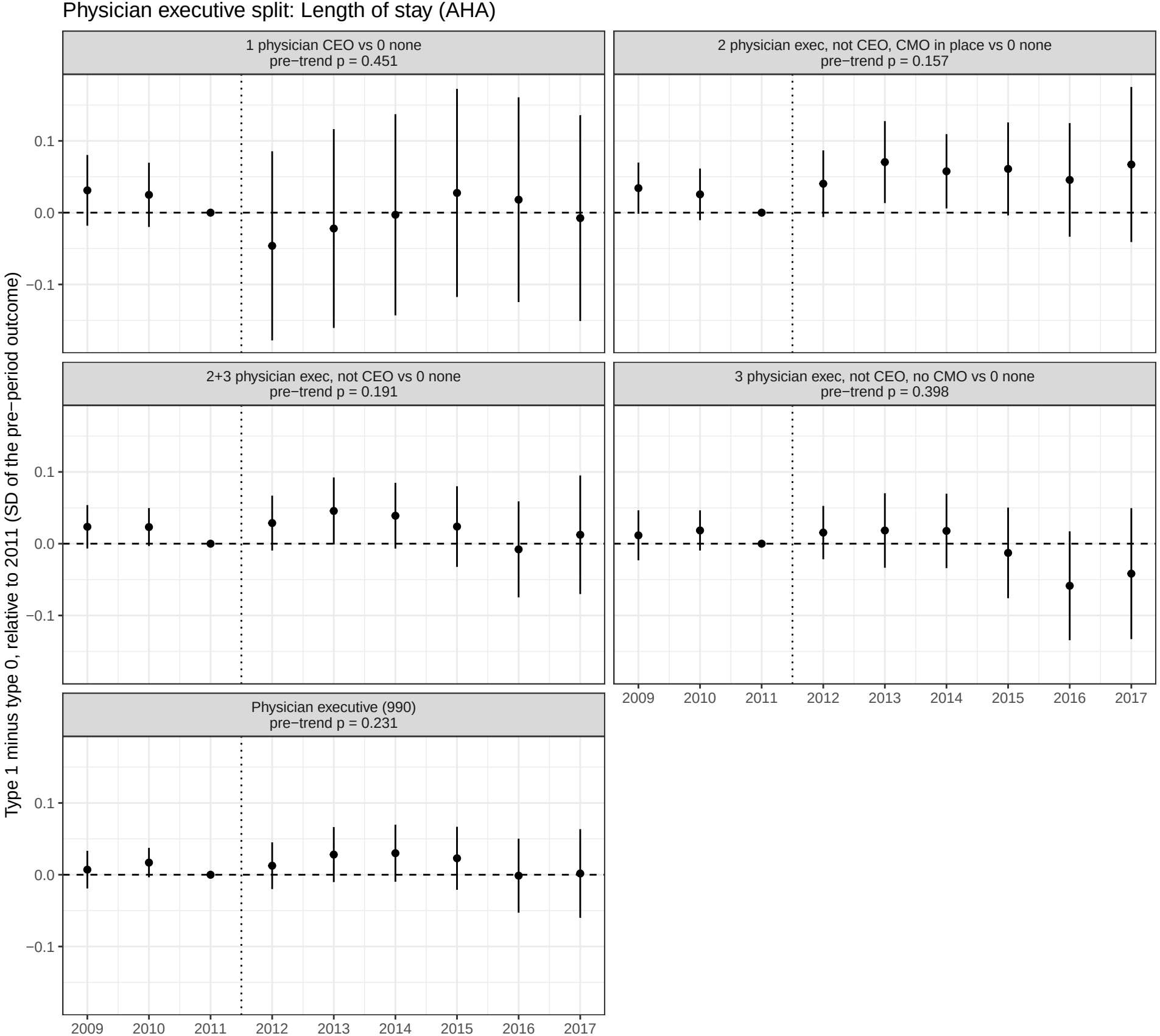

Physician executive split: log equipment purchases
1 physician CEO vs 0 none
pre−trend p = 0
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.412
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.219
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.141
Physician executive (990)
pre−trend p = 0.132
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0
−2
−4
−6
−8
2009
2010
2011
2012
2013
2014
2015
2016
2017

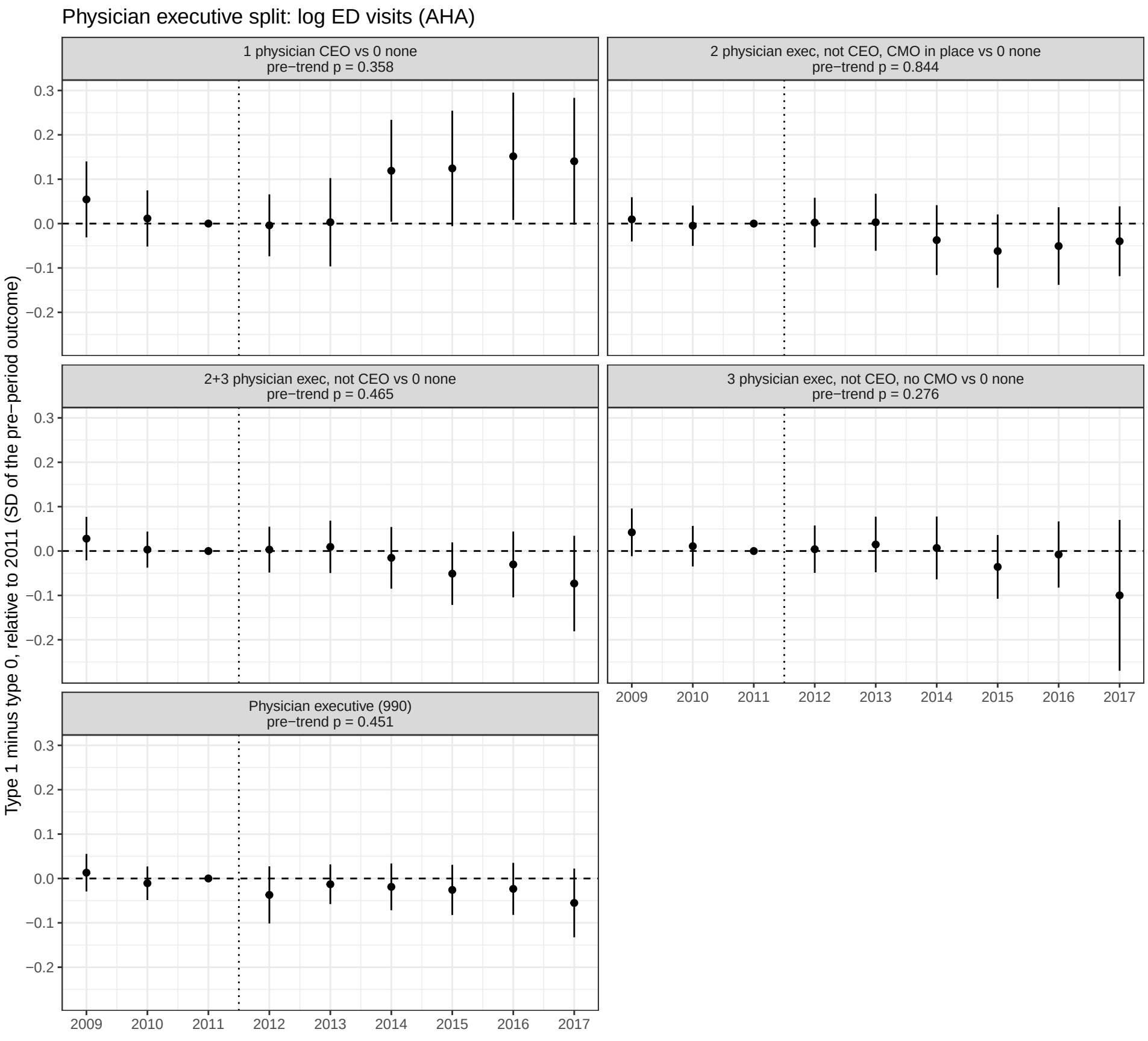

Physician executive split: log total discharges
1 physician CEO vs 0 none
pre−trend p = 0.416
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.425
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.384
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.406
Physician executive (990)
pre−trend p = 0.61
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.1
0.0
−0.1
2009
2010
2011
2012
2013
2014
2015
2016
2017

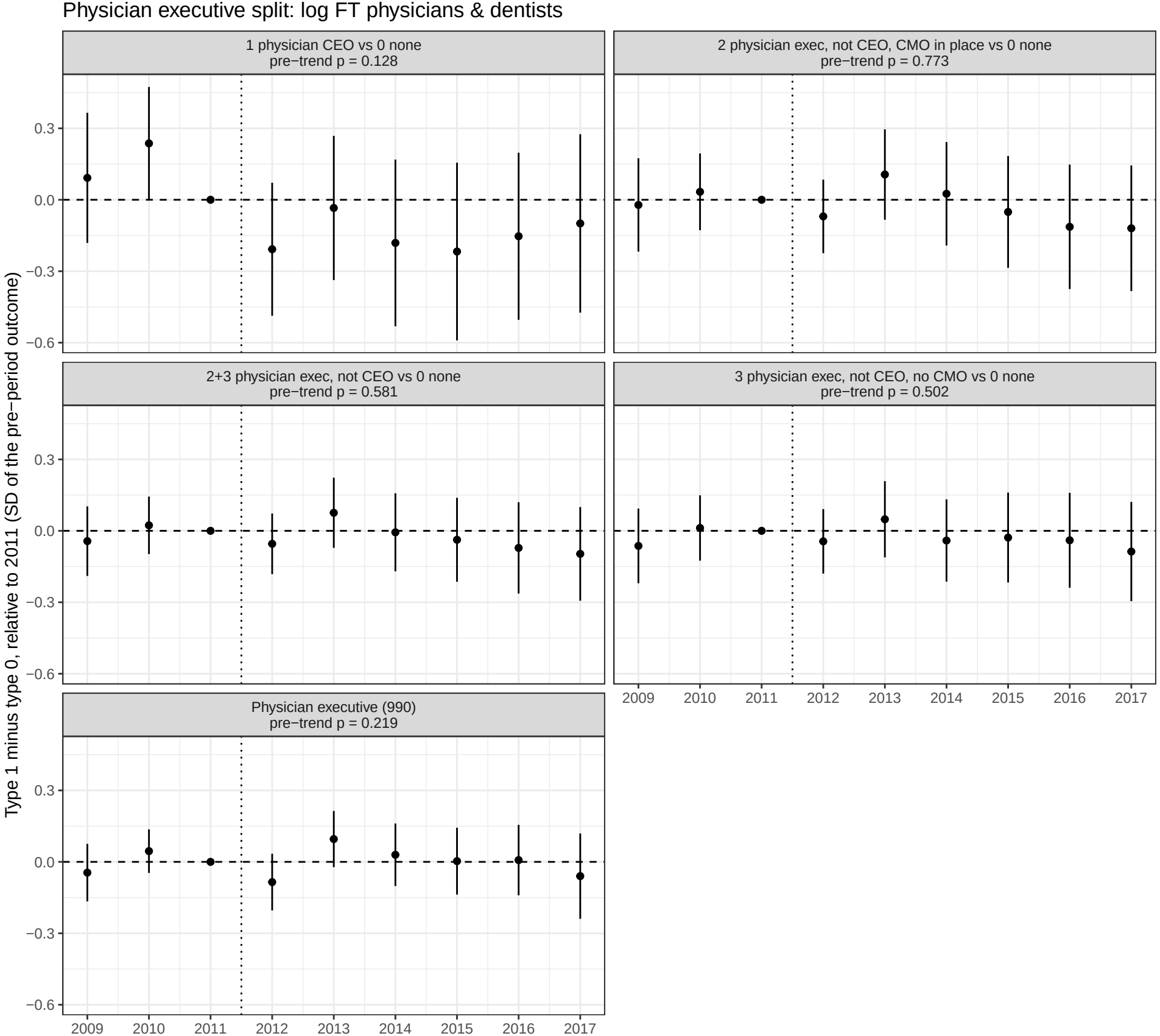

Physician executive split: Medicare share of discharges
1 physician CEO vs 0 none
pre−trend p = 0.951
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.963
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.965
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.965
Physician executive (990)
pre−trend p = 0.937
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.1
0.0
−0.1
−0.2
2009
2010
2011
2012
2013
2014
2015
2016
2017

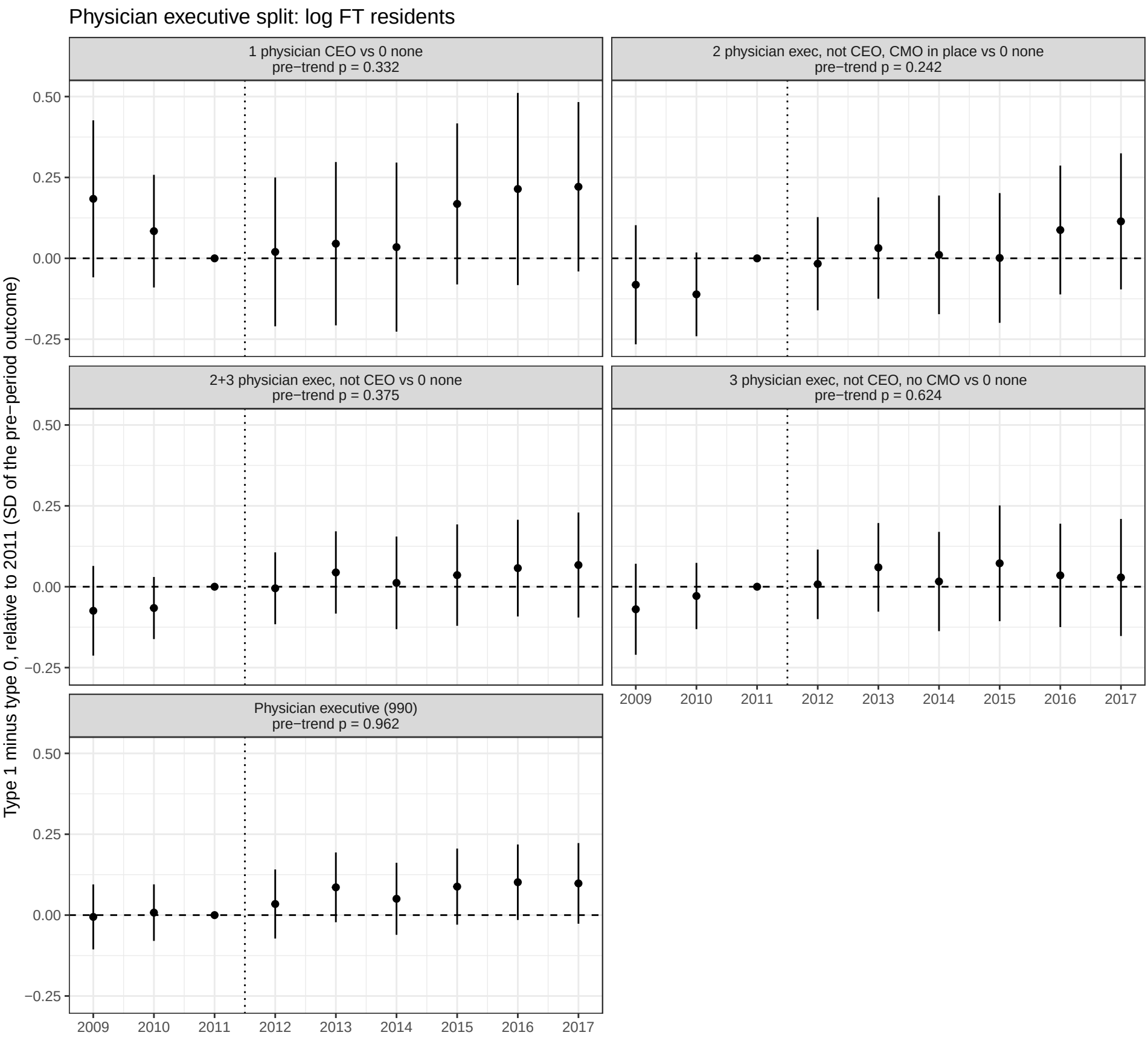

Physician executive split: Medicaid share of discharges
1 physician CEO vs 0 none
pre−trend p = 0.28
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.985
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.918
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.747
Physician executive (990)
pre−trend p = 0.899
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.4
0.2
0.0
−0.2
−0.4
2009
2010
2011
2012
2013
2014
2015
2016
2017

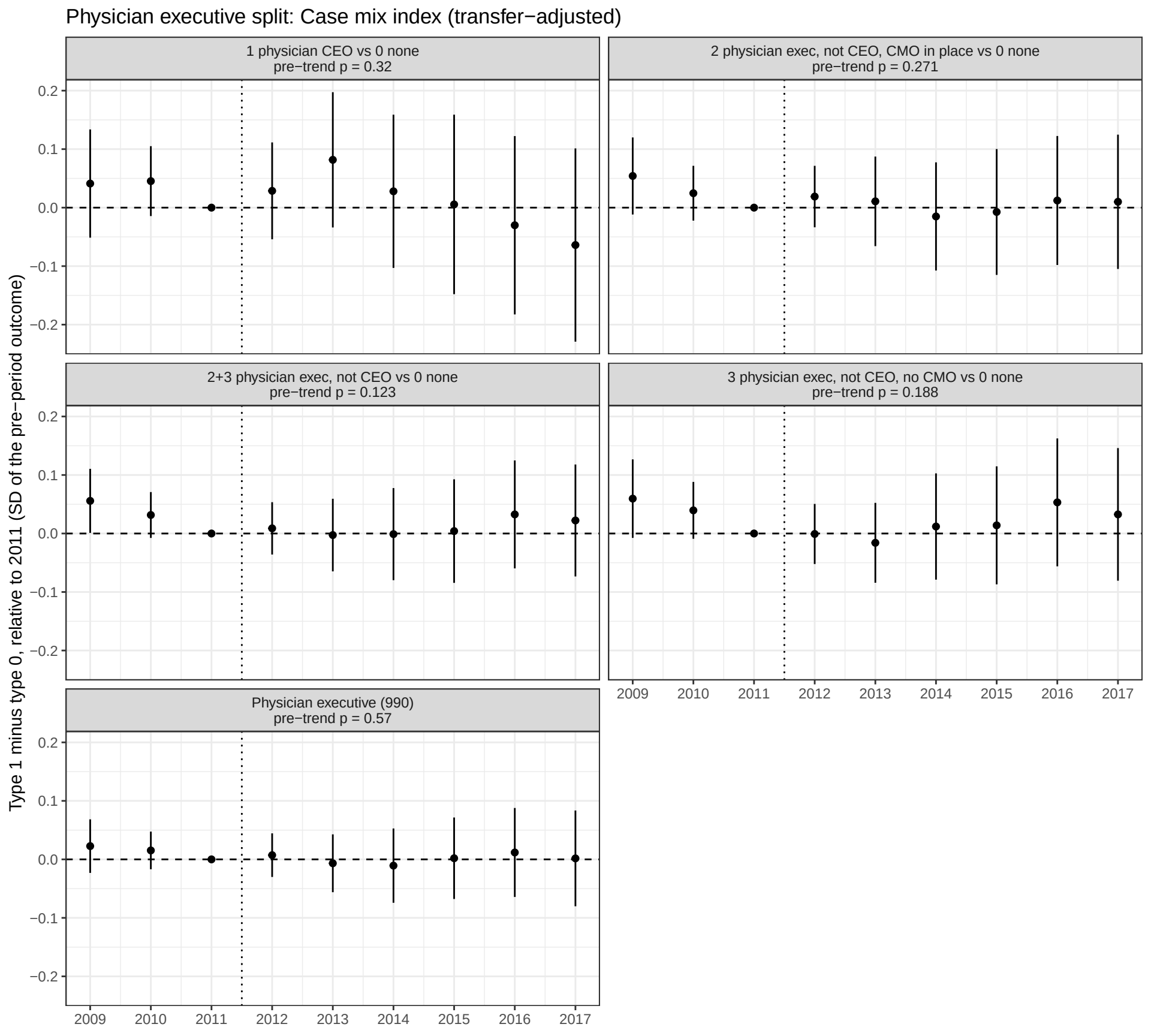

Physician executive split: log admissions (AHA)
1 physician CEO vs 0 none
pre−trend p = 0.183
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.193
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.049
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.085
Physician executive (990)
pre−trend p = 0.312
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.05
0.00
−0.05
2009
2010
2011
2012
2013
2014
2015
2016
2017

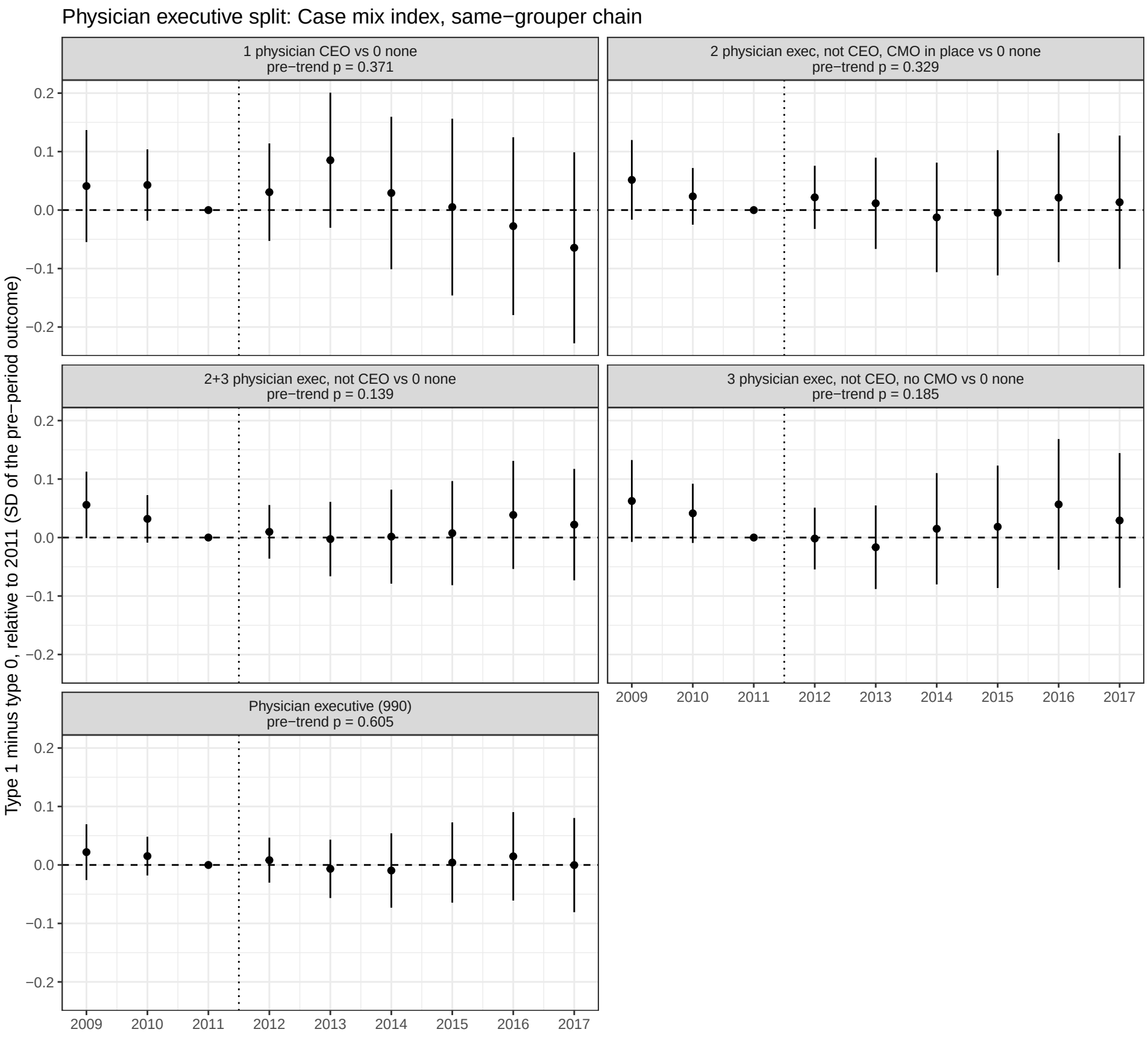

Physician executive split: Length of stay (AHA)
1 physician CEO vs 0 none
pre−trend p = 0.451
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.157
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.191
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.398
Physician executive (990)
pre−trend p = 0.231
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.1
0.0
−0.1
2009
2010
2011
2012
2013
2014
2015
2016
2017

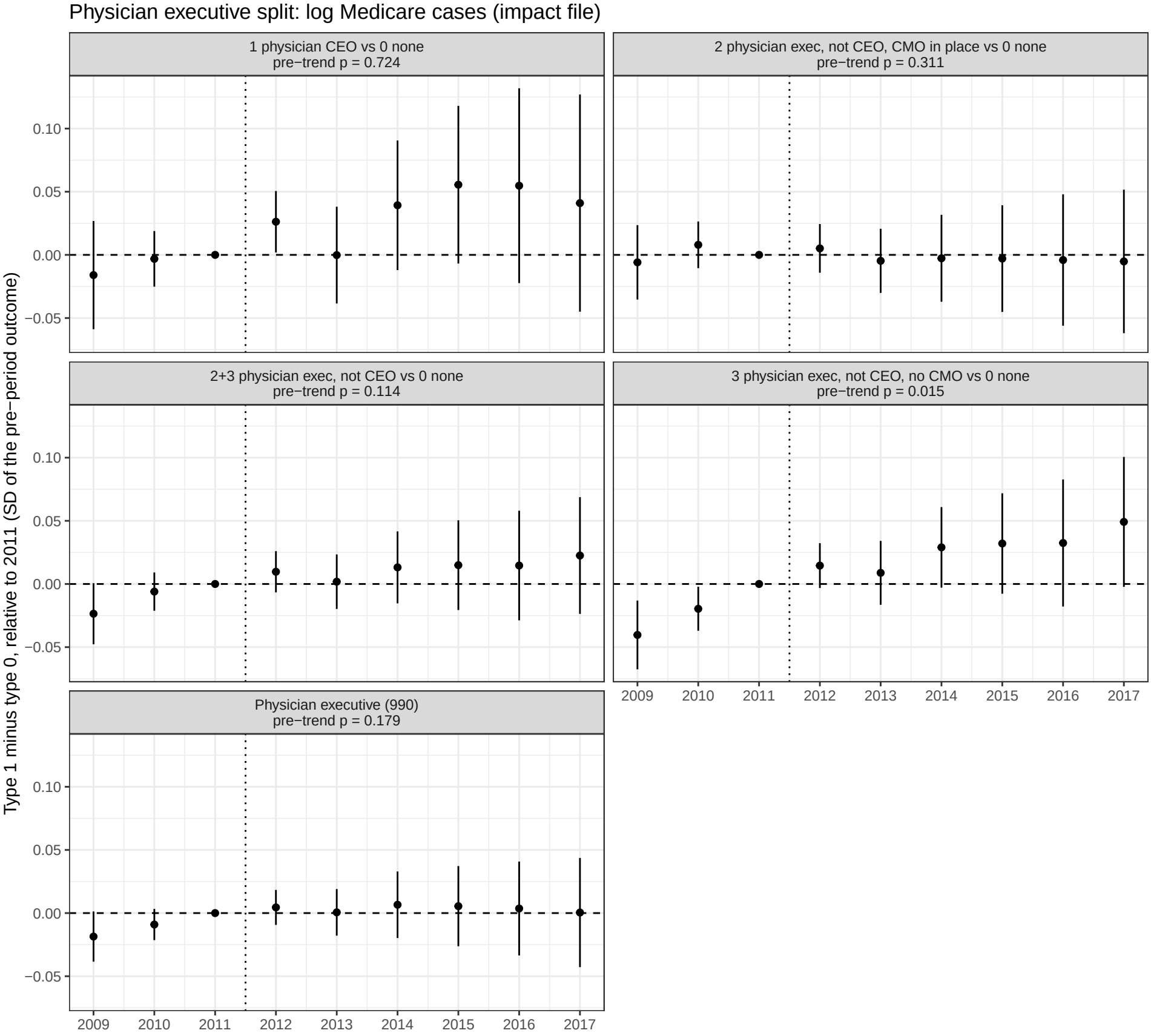

Physician executive split: log ED visits (AHA)
1 physician CEO vs 0 none
pre−trend p = 0.358
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.844
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.465
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.276
Physician executive (990)
pre−trend p = 0.451
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.3
0.2
0.1
0.0
−0.1
−0.2
2009
2010
2011
2012
2013
2014
2015
2016
2017

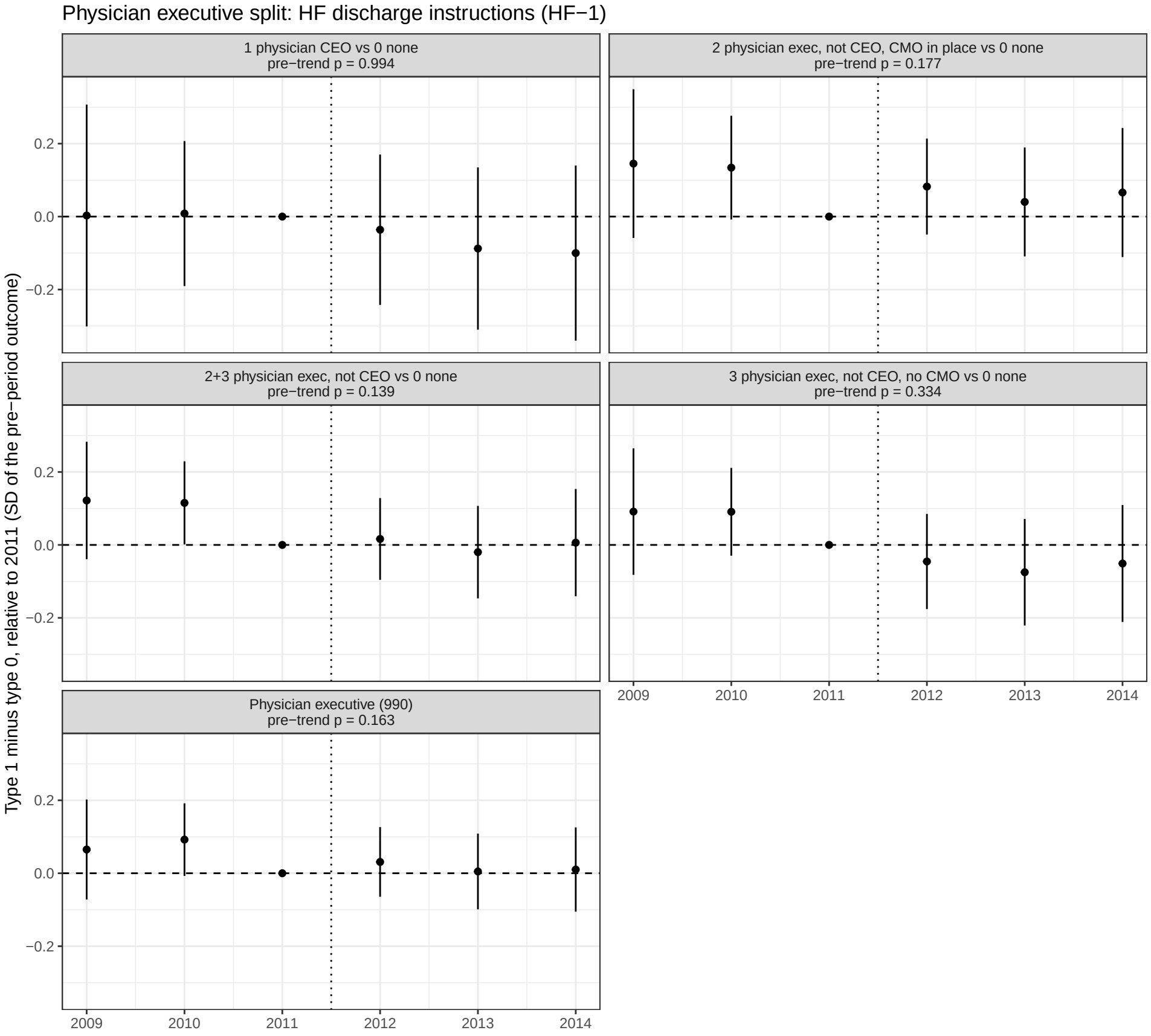

Physician executive split: log FT physicians & dentists
1 physician CEO vs 0 none
pre−trend p = 0.128
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.773
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.581
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.502
Physician executive (990)
pre−trend p = 0.219
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.3
0.0
−0.3
−0.6
2009
2010
2011
2012
2013
2014
2015
2016
2017

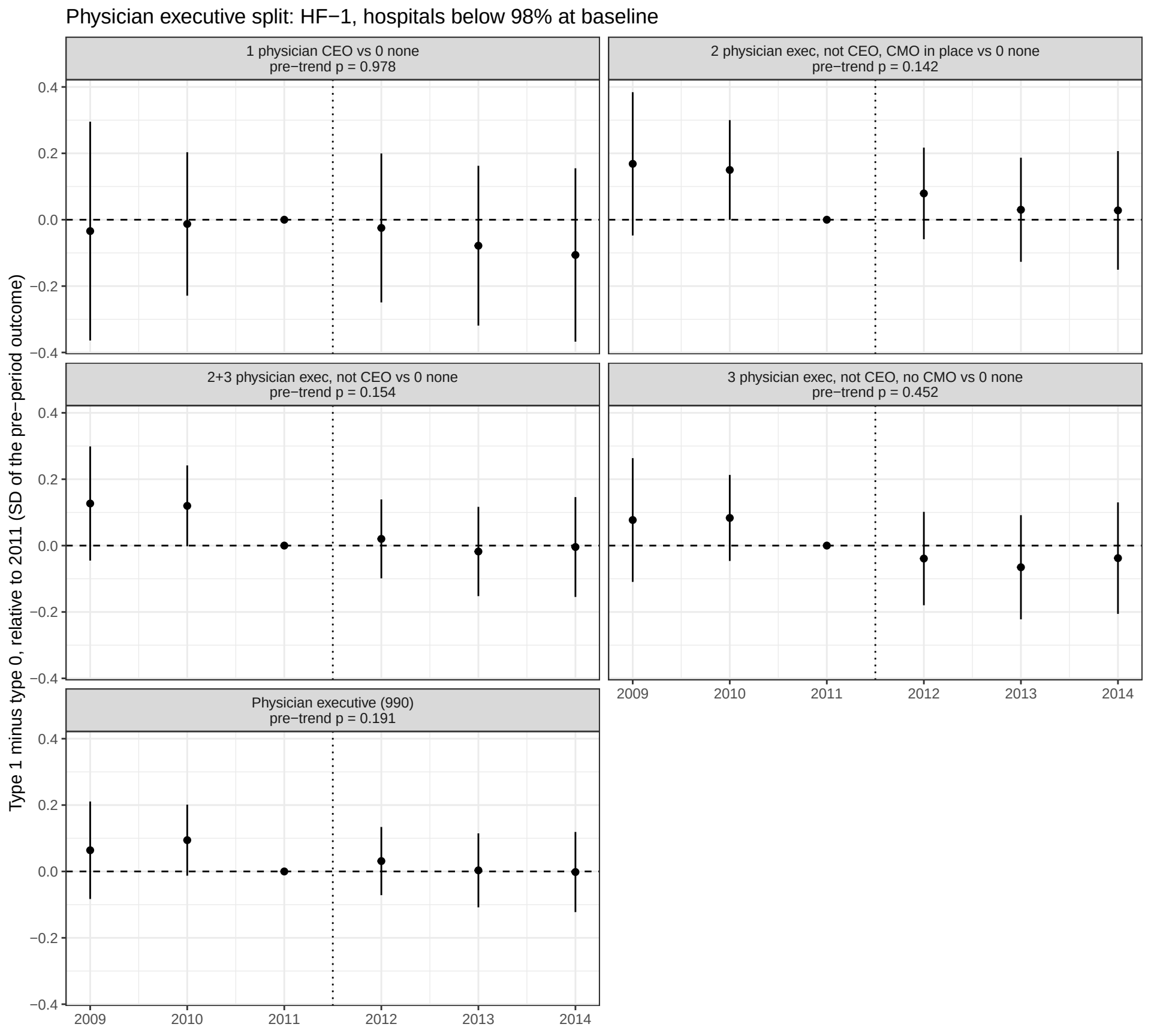

Physician executive split: log FT residents
1 physician CEO vs 0 none
pre−trend p = 0.332
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.242
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.375
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.624
Physician executive (990)
pre−trend p = 0.962
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.50
0.25
0.00
−0.25
2009
2010
2011
2012
2013
2014
2015
2016
2017

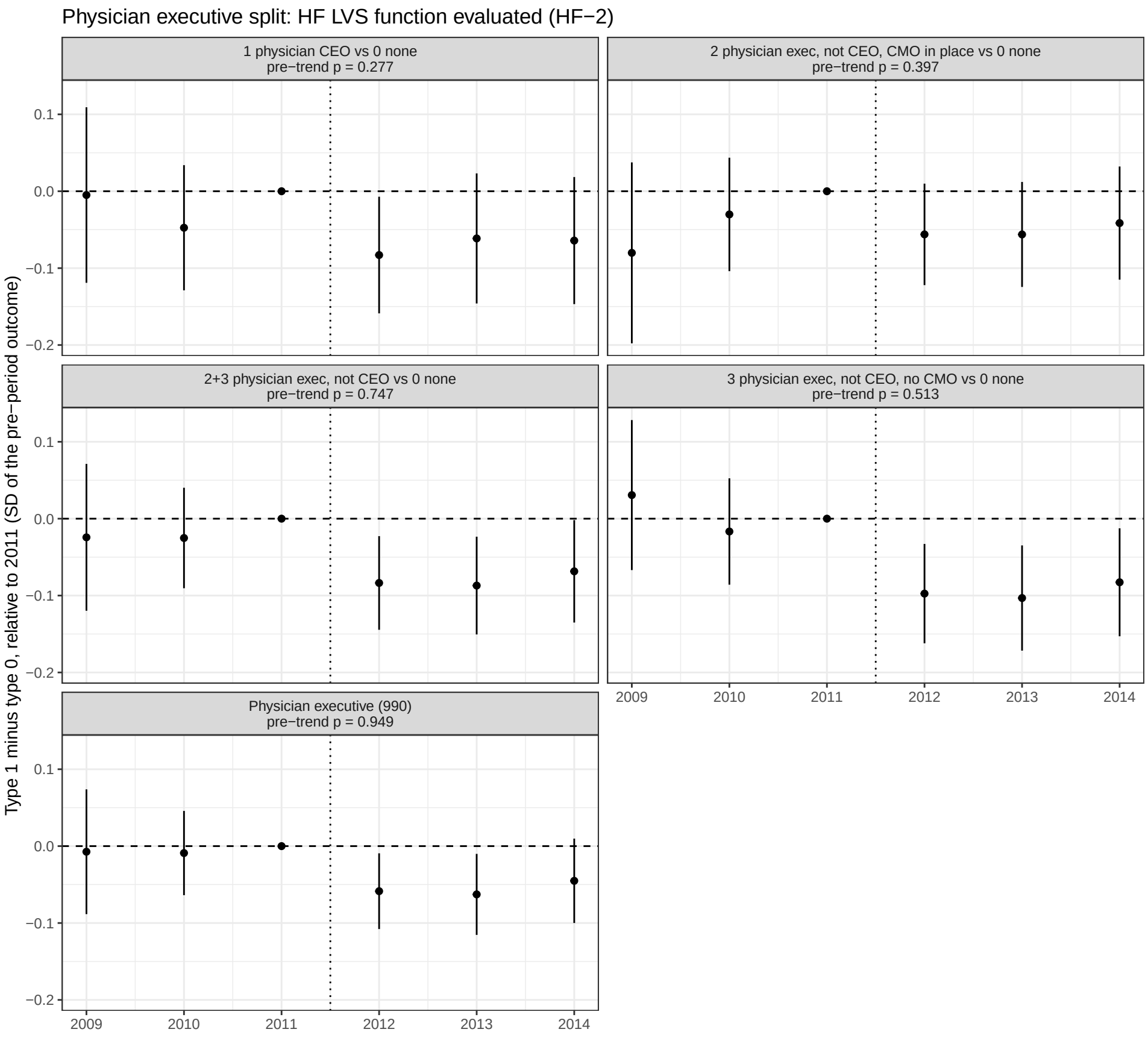

Physician executive split: Case mix index (transfer−adjusted)
1 physician CEO vs 0 none
pre−trend p = 0.32
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.271
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.123
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.188
Physician executive (990)
pre−trend p = 0.57
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.1
0.0
−0.1
−0.2
2009
2010
2011
2012
2013
2014
2015
2016
2017

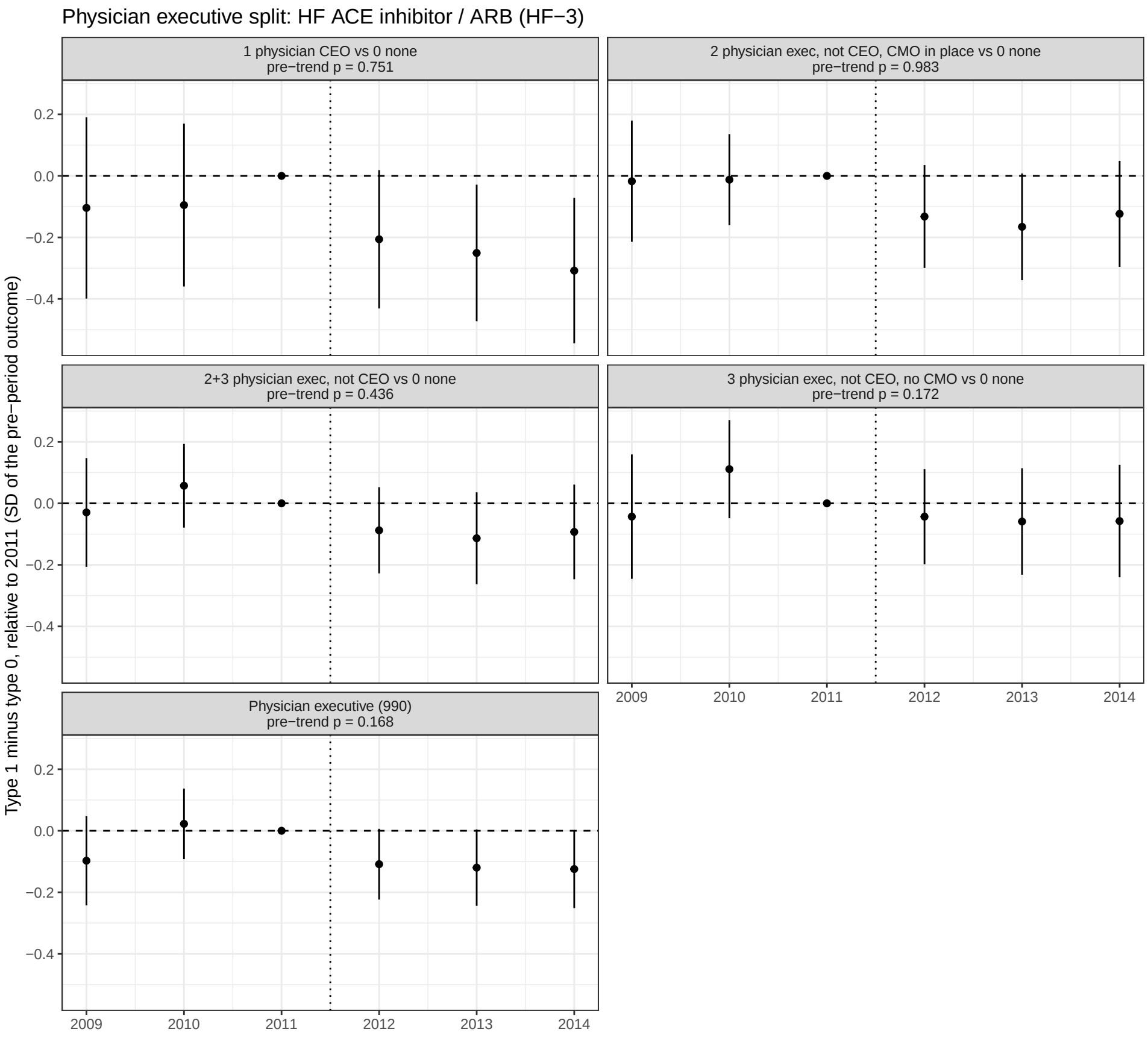

Physician executive split: Case mix index, same−grouper chain
1 physician CEO vs 0 none
pre−trend p = 0.371
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.329
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.139
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.185
Physician executive (990)
pre−trend p = 0.605
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.1
0.0
−0.1
−0.2
2009
2010
2011
2012
2013
2014
2015
2016
2017

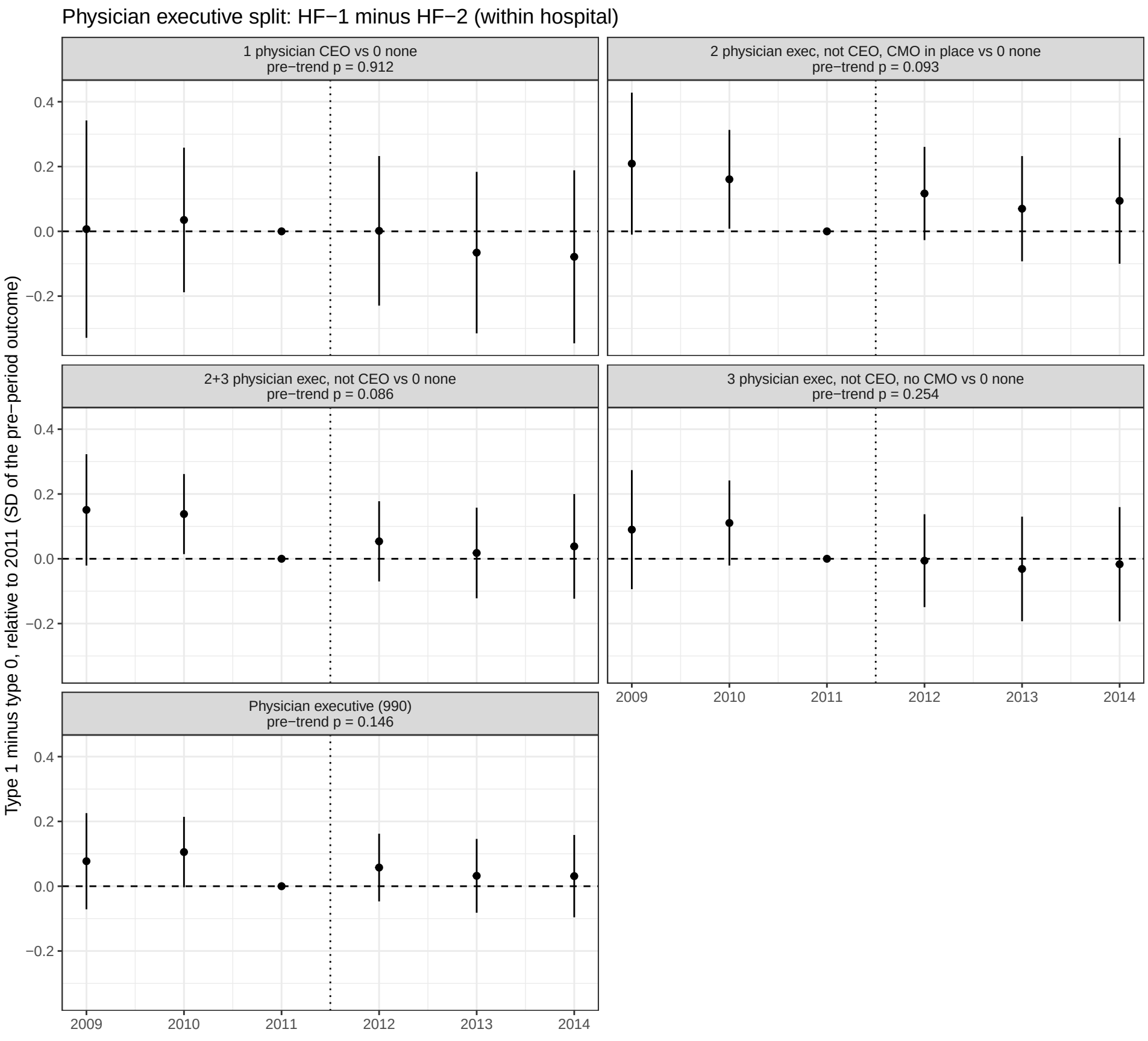

Physician executive split: log Medicare cases (impact file)
1 physician CEO vs 0 none
pre−trend p = 0.724
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.311
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.114
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.015
Physician executive (990)
pre−trend p = 0.179
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.10
0.05
0.00
−0.05
2009
2010
2011
2012
2013
2014
2015
2016
2017

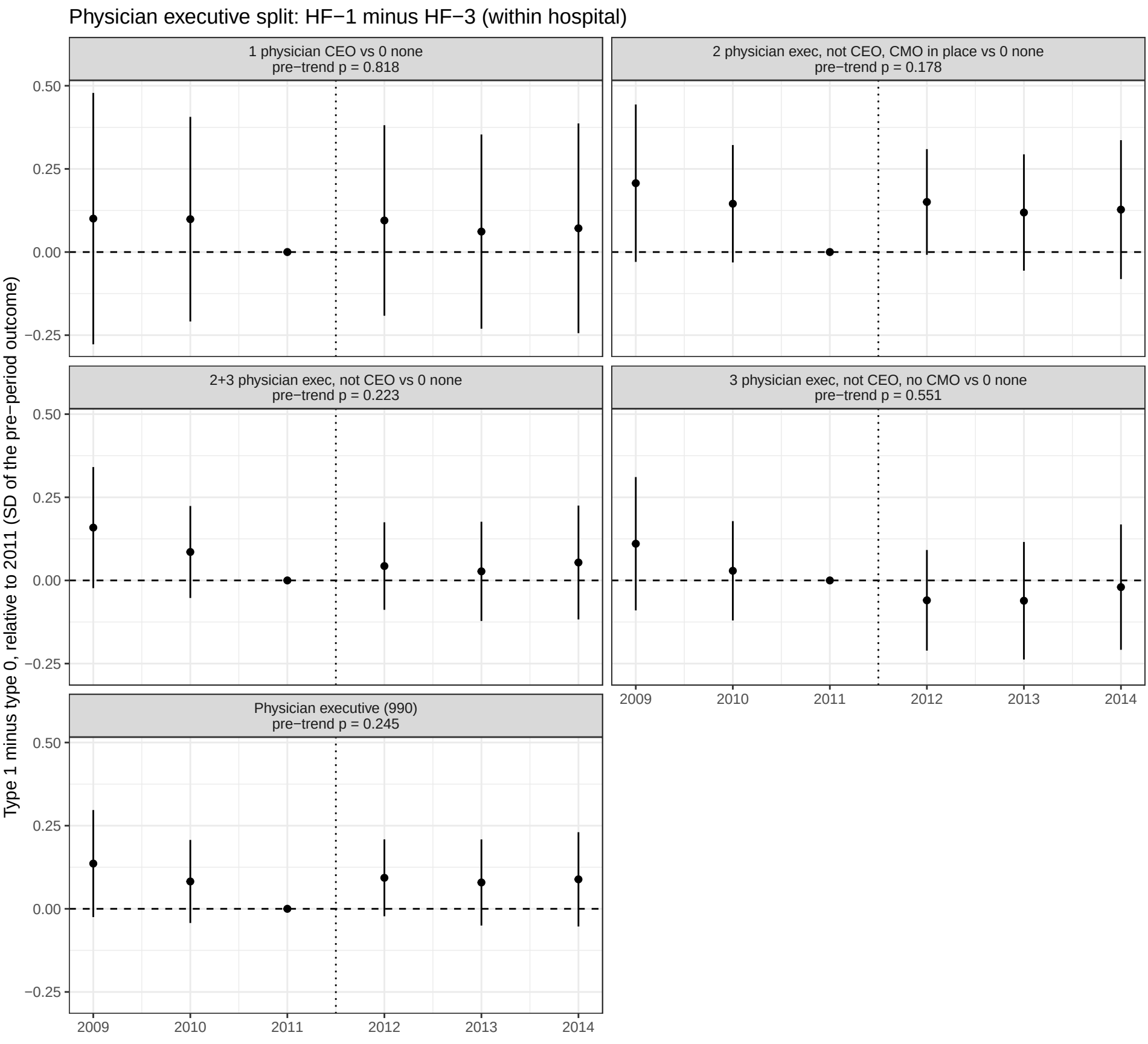

Physician executive split: HF discharge instructions (HF−1)
1 physician CEO vs 0 none
pre−trend p = 0.994
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.177
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.139
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.334
Physician executive (990)
pre−trend p = 0.163
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.0
−0.2
2009
2010
2011
2012
2013
2014


Physician executive split: HF−1, hospitals below 98% at baseline
1 physician CEO vs 0 none
pre−trend p = 0.978
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.142
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.154
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.452
Physician executive (990)
pre−trend p = 0.191
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.4
0.2
0.0
−0.2
−0.4
2009
2010
2011
2012
2013
2014


Physician executive split: HF LVS function evaluated (HF−2)
1 physician CEO vs 0 none
pre−trend p = 0.277
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.397
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.747
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.513
Physician executive (990)
pre−trend p = 0.949
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.1
0.0
−0.1
−0.2
2009
2010
2011
2012
2013
2014


Physician executive split: HF ACE inhibitor / ARB (HF−3)
1 physician CEO vs 0 none
pre−trend p = 0.751
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.983
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.436
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.172
Physician executive (990)
pre−trend p = 0.168
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.2
0.0
−0.2
−0.4
2009
2010
2011
2012
2013
2014


Physician executive split: HF−1 minus HF−2 (within hospital)
1 physician CEO vs 0 none
pre−trend p = 0.912
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.093
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.086
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.254
Physician executive (990)
pre−trend p = 0.146
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.4
0.2
0.0
−0.2
2009
2010
2011
2012
2013
2014


Physician executive split: HF−1 minus HF−3 (within hospital)
1 physician CEO vs 0 none
pre−trend p = 0.818
2 physician exec, not CEO, CMO in place vs 0 none
pre−trend p = 0.178
2+3 physician exec, not CEO vs 0 none
pre−trend p = 0.223
3 physician exec, not CEO, no CMO vs 0 none
pre−trend p = 0.551
Physician executive (990)
pre−trend p = 0.245
Type 1 minus type 0, relative to 2011 (SD of the pre−period outcome)
0.50
0.25
0.00
−0.25
2009
2010
2011
2012
2013
2014